

CHAPTER 4

Microturbine Generation Power Systems

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4.1 INTRODUCTION

There are various methods to produce energy such as wind turbine, solar cell, microturbine (MT), and fuel cell but MT offers several potential advantages compared to other technologies for small-scale power generation.

Generally, MT is a small gas turbine, which has similar cycles and in many cases its components are made of the same materials as a heavy gas turbine. The MT power-to-weight ratio is superior to the heavy gas turbine because based on aerodynamic equations, reduction of turbine diameters cause an increase in shaft rotational speed, so output power is improved. Heavy gas turbine generators are too large and too expensive for distributed power applications, so, MTs are developed for small-scale power like electrical power generation alone or as combined cooling, heating, and power (CCHP) systems.

High power-to-weight ratio and efficiency are not the only advantages of microturbine generation (MTG) systems. Advantages that make MTs superior to other distributed generation (DG) systems are compact size, high reliability, light weight, low operation and maintenance costs, simple and remote control, low environmental emissions, fewer moving parts, low noise level, and fuel flexibility [1]. MTs can operate on a variety of gaseous or liquid fuels including natural gas, associated gas, LPG/propane, flare gas, landfill gas, digester gas, diesel, biogas, and kerosene, some of which are renewable energy.

The interest in MTGs is increasing due to technical, economic, and reliability merits. Potential applications of MTs include peak shaving, premium power, remote power, cogeneration applications, and as grid supporter.

This chapter presents applications, configurations, and operational studies of MTs as DG. First, the MT and its operation cycle are introduced and power electronic interface and type of MT are discussed. Finally, dynamic modeling of MT is considered and simulation study is carried out in PSCAD/EMTDC software for grid-connected and isolated operation modes.

4.2 HISTORY

The idea of using a turbine was hatched many years ago, and Leonardo da Vinci was the first man who drew an illustration of his idea, which he called a “chimney jack.” Early models were based on the steam turbine. The first true gas turbine was invented by John Barber in 1791, and his invention had most of the elements present in modern gas turbines. Over the years, many gas turbine designs were based on John Barber’s model. In fact, gas turbines were involved in many military and aerospace industrial projects done during the 1950s through the 1970s. Using the gas turbine as a constant power producer was done in the 1980s and 1990s and the concept of DG was accelerated.

In 1950s and 1960s, the Boeing Company was a pioneer in using gas turbines for helicopters, aircraft, and army tanks. The first cogeneration based on a gas turbine was installed by California Gas Company in 1962. The Boeing Company developed a 100 kW outboard motor that had positive effects like reduction of weight and fuel of gas turbines, but the company failed to keep its products and, finally, this product line was sold to Caterpillar. In 1960s, big companies like BMW, Rover, Nissan, Ford, and Benz produced gas turbine-based automobiles and Rover pioneered them.

In 1980, a program entitled Advanced Energy Systems supported by the Gas Research Institute was initiated to develop a small gas turbine rated at 50 kW and including heat recovery. In 1990, this program was abandoned due to problems of product final cost. Since then, Gas Research Institute has supported new projects with other companies like Northern Research & Engineering Energy System and NoMac Energy Systems.

In 1988, Robin Mackay and Jim Noe, former members of Allied Signal Company and Garrett, respectively, decided to establish a new company, NoMac Energy Systems, for developing a small gas turbine in the automotive industry. In fact, NoMac was the first manufacturer of MTs in the world, under the support of the Gas Research Institute. In 1993, the company name was changed to Capstone Turbine Corporation, which is one of the biggest MT manufacturers in the world.

Generally, research on turbochargers in the transportation industry and the aerospace industry as well as improvement of gas turbine efficiency were the main reasons to develop MTs. Recently, output power of MTs has reached up to 500 kW and improvement of efficiency and size is considered in new models.

4.3 STATISTICAL INFORMATION

In this section, MT statistical aspects are considered and charts are collected from credible sources. There are various methods to generate electricity, which are subsets of renewable sources or fossil fuels. Fossil fuels are the traditional method and now the international focus is on renewable energies. There are several factors to consider when choosing a power source such as pollutant emission level, fuel cost, the cost of construction, and availability. Global warming and greenhouse gasses have driven trends toward renewable energy, but these energies (e.g., wind power or solar power) are not reliable resources and they are not the main sources for power generation. Fig. 4.1 shows that along with the increase of energy demand, the total share of fossil fuels will not be affected by a drastic reduction. With respect to growth of renewable energies, there is a gap between them and fossil fuels to be the main resource of power generation.

The use of fossil fuels from 1965 to 2035 by percentage is shown in Fig. 4.2.

Fossil fuels share of the primary energy mix will decrease only modestly while the new renewable share will increase. Fossil fuels will remain the

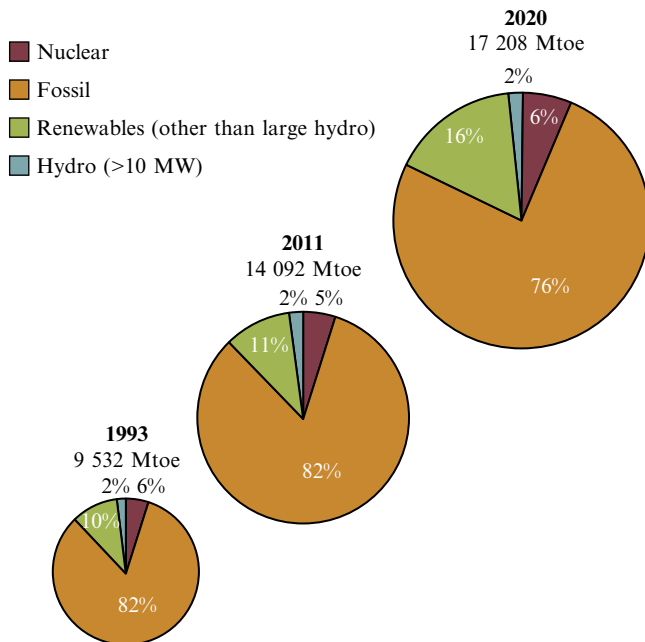


Fig. 4.1 Total primary energy supply by resource 1993, 2011, and 2020 [2].

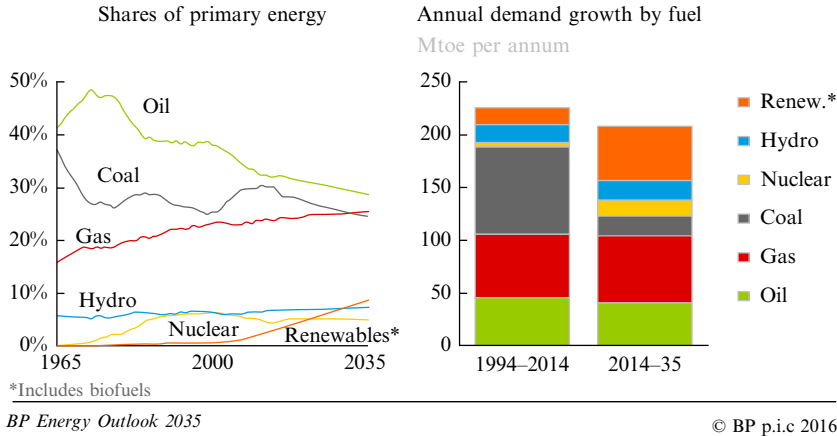


Fig. 4.2 Evolution of world primary energy consumption [3]. *Used with permission from 2016 BP Energy outlook 2035.*

dominant source of energy powering the global economy, providing around 60% of the growth in energy and accounting for almost 80% of total energy supply in 2035. Gas is the fastest growing fossil fuel (1.8% p.a.), with its share in primary energy gradually increasing [3].

Fig. 4.3 presents percentage shares of energy sources with greater clarity. Global interest in natural gas is obviously increasing during the 1975–2035 period.

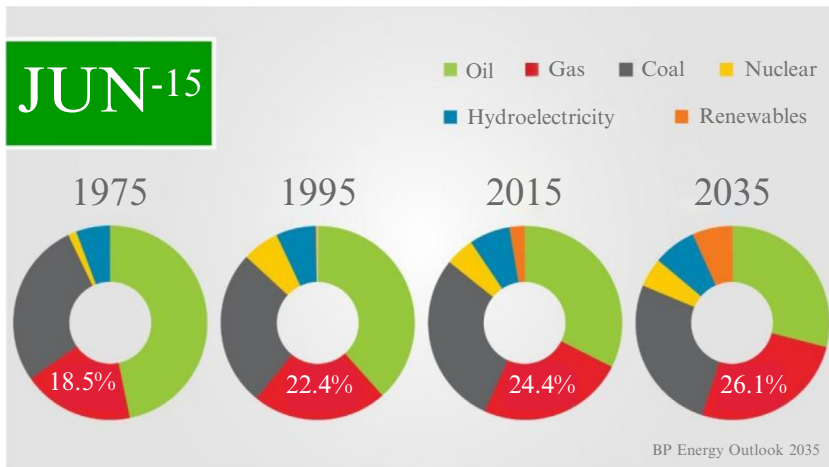


Fig. 4.3 Evolution of natural gas in power generation [3]. *Used with permission from <http://www.bp.com/en/global/corporate/bp-magazine/observations/today-and-tomorrow-the-rise-of-gas.html>.*

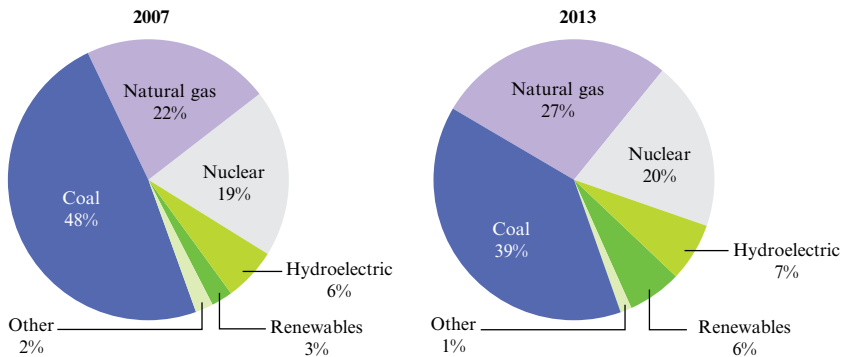
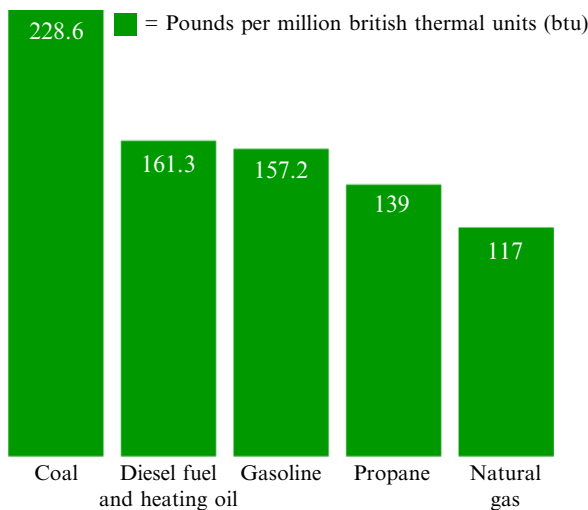


Fig. 4.4 Changes in US electricity generation mix, 2007–13 [4].

The United States is one of the highest power consuming countries in the world that has accordant growth in the use of natural gas. Fig. 4.4 shows percentages of US electricity generation in different years. As coal generation declines, it has been replaced by generation from a combination of cleaner power sources led by natural gas [4].

The use of natural gas is increasing due to high reliability and clean fuel. The chart in Fig. 4.5 compares the amounts of carbon dioxide emitted per unit (p.u.) of heat content generated. As shown, natural gas is the cleanest fossil fuel.



Source: US Energy Information Administration

Fig. 4.5 Amounts of carbon dioxide emitted by fossil fuels [3]. Used with permission from <http://www.bp.com/en/global/corporate/bp-magazine/observations/today-and-tomorrow-the-rise-of-gas.html>.

Monthly and annual average natural gas spot price at henry hub (1996–2015)
dollars per million British thermal unit

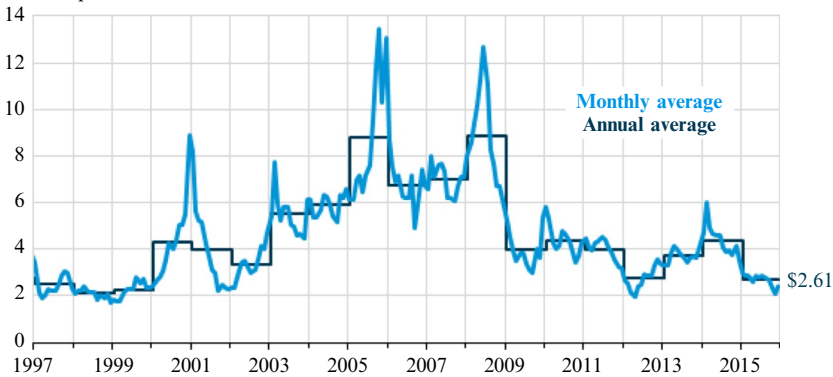


Fig. 4.6 Natural gas prices, 1997 through 2015 [4].

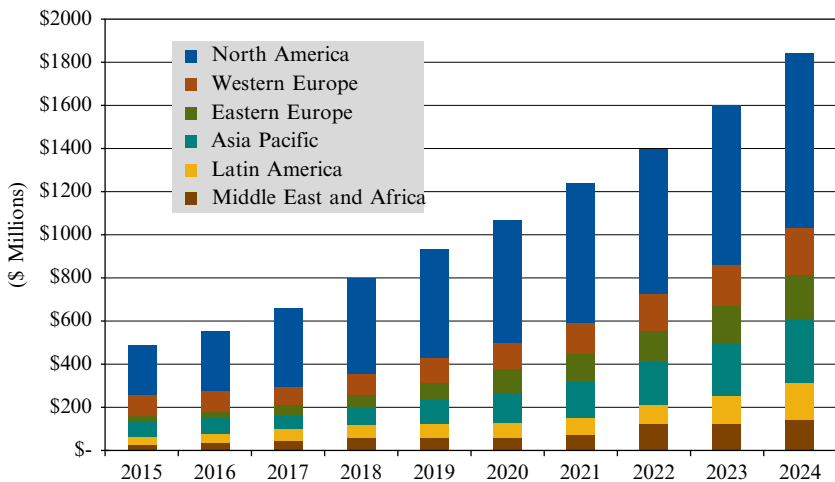
The natural gas industry has a well-documented history of price volatility. Due to the nature of its production, storage, and transmission; the fact that it supplies many end uses; and its susceptibility to extreme weather events, the price of natural gas fluctuates more than does the price of its competitors. For example, the natural gas market experienced a number of dramatic price swings between 2000 and 2014, which is shown in Fig. 4.6.

In 2005, the spot market price spiked to nearly \$14 per million British thermal unit (MMBtu) in response to hurricane activity in the Gulf of Mexico, where many gas wells are located. In contrast, prices dipped to about \$2 per MMBtu in 2012 in response to decreased demand resulting from the economic recession and from a series of warm winters, which reduced competition between heating and electricity suppliers for limited natural gas pipeline resources [5].

Many experts believe that low natural gas prices are not sustainable over the long term. For example, the US Energy Information Administration's (EIA) Annual Energy Outlook projects that spot prices will significantly increase from the recent low point of \$2.75 per MMBtu in 2012 to \$6.03 per MMBtu in 2030 and \$7.65 per MMBtu in 2040. Factors that contribute to upward pressure on prices and the risk of price volatility include uncertain available supply and potentially increasing demand for natural gas from electric utilities, other competing domestic users, and exporters [5].

The market for new MT installations is expected to surpass \$1 billion in annual revenue by 2020. Over the next decade, Navigant Research projects more than 4 GW of cumulative MT capacity will be installed globally. Despite this healthy growth, the crowded DG technology landscape is

expected to temper upside growth for MT installations over the forecast period. North America and Asia Pacific are expected to account for 63% of the cumulative revenue generated from global MT installments between 2015 and 2024. Substantial growth in shale gas production and an improving combined heat and power (CHP) environment point to the North American market leading all regions with \$5.2 billion in cumulative revenue. This represents nearly half of the cumulative revenue generated globally. Annual MT revenue in North America is expected to experience a CAGR¹ of 14.8%. Asia Pacific is a distant second to the North American market with \$1.4 billion in cumulative revenue generated from MT installments, but the region is expected to experience higher annual revenue growth between 2015 and 2024 (CAGR of 17.7%). Western Europe is third with \$1.3 billion in cumulative revenue, followed by Eastern Europe at \$1.0 billion [6]. Fig. 4.7 shows annual MT revenue



(Source: Navigant Research)

Fig. 4.7 Annual microturbine revenue by region, world markets: 2015–24 [6]. From *Microturbines: Global Market Analysis and Forecasts for Residential, Commercial, and Industrial Applications*. Navigant research. From <http://www.prnewswire.com/news-releases/microturbines-global-market-analysis-and-forecasts-for-residential-commercial-and-industrial-applications-300105652.html>.

¹ CAGR refers to compound average annual growth rate, using the formula: $CAGR = (\text{End Year Value} \div \text{Start Year Value})^{(1/\text{steps})} - 1$. CAGRs presented in the tables are for the entire timeframe in the title. Where data for fewer years are given, the CAGR is for the range presented. Where relevant, CAGRs for shorter timeframes may be given as well.

by region and represents increasing global MTG market size as a reliable resource. MTs can operate with various fuels in addition to natural gas. Fuel flexibility is one of the favorable factors influencing the choice of MTs.

4.4 GAS TURBINE

Today, gas turbines play an important role in electric power systems. Use of the gas turbine has increased in aircraft propulsion, the automotive industry, and the oil and gas industry for driving pumps and other uses. There are many reasons that distinguish gas turbines from other sources. One of most striking features of a gas power plant is fast startup time, which is suitable for peak shaving and used as a base load in electric power systems. High power to weight ratio, long life, high reliability, and remote control are other advantages of gas turbines. Compared with other plants such as thermal plants, this system needs low water cooling and therefore water is not a restriction on the construction of a gas plant.

A gas turbine is a rotating engine that extracts energy from a flow of combustion gases (resulting from the ignition of compressed air and a fuel-like natural gas). Gas turbines consist of three general parts: a compressor, a combustor, and a turbine. A basic condition for turbines' power generation is the difference between inlet pressure and outlet pressure, so the compressor is used to increase inlet pressure and the compressor is located before the turbine. Compressed air cannot produce enough mechanical power for driving a turbine because the compressor consumes a large part of turbine output power. For driving a turbine, more energy is required and a fuel like natural gas is needed. Compressed air and fuels are mixed in the combustor due to more power production. Therefore, the combustor or combustion chamber is located between the compressor and the turbine. Fig. 4.8 shows a gas turbine diagram. Gas turbine operation is based on the Brayton cycle that will be described in detail below.

Output efficiency of gas turbine depends on compressor and turbine efficiency as well as environmental conditions like temperature. For improving output efficiency, high-temperature inlet gas is necessary. In general, the electrical efficiencies of modern gas turbines range from 30% to 40% and output power is about 1–500 MW.

There are several type of gas turbine such as turboprop, MT, and turbofan and the purpose of this chapter is the detailed description of MTs.

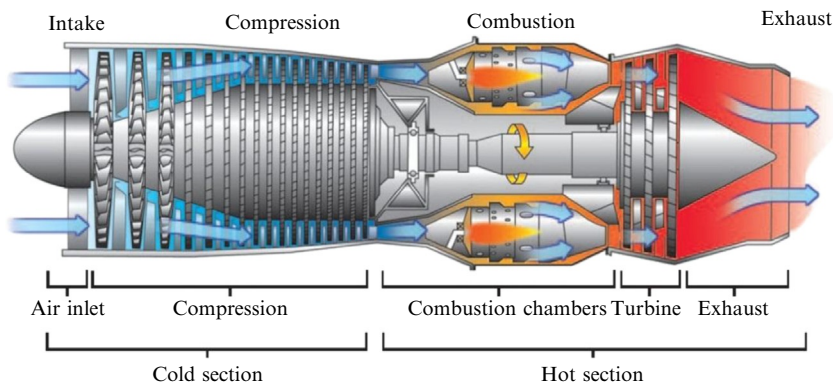


Fig. 4.8 Gas turbine generation system [7].

4.4.1 The Brayton Cycle

Large gas turbines operations are based on the Brayton thermodynamic cycle. Understanding of the Brayton cycle concept is an essential issue because MTs are a subset of gas turbines and they have a similar cycle.

This cycle was proposed by George Brayton (1830–92), who was an American engineer, in 1872. He applied this cycle for his patent that was for a reciprocating oil-burning engine. Today, gas turbines and jet engines based on the Brayton cycle are widely used in industry.

The Brayton cycle or Joule cycle is a thermodynamic cycle and needs to have a working fluid. Working fluid in a thermodynamic system is a liquid or gas that absorbs or transmits. Air is the working fluid of the Brayton cycle. This system is based on four steps: compression, heat addition, expansion, and heat rejection, and each cycle consists of two isobaric processes and two adiabatic processes. An isobaric process is a thermodynamic process in which pressure stays constant and an adiabatic process is a thermodynamic process without heat transferring between two systems.

In a Brayton open cycle, inlet fresh air at ambient conditions is running through a compressor and after that, the resulting compressed air enters the combustion chamber. In the combustor, fuel is added to air and ignited, which causes an increase of gases volume. Expanded gases run through a turbine and provide mechanical force for driving. Finally, combustion products are discharged to the atmosphere. Fig. 4.9 diagrams the open cycle that discharges air flow to the atmosphere and occurs in each cycle. The compressor

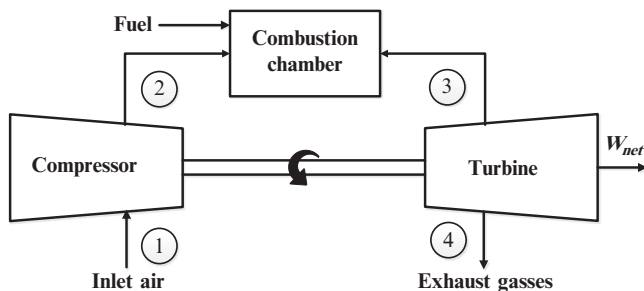


Fig. 4.9 An open cycle gas turbine engine.

power requirement is about 40%–80% of turbine power output, which is a large amount, so power loss reduction is necessary.

A typical close cycle and T-S and P-V diagram of an ideal Brayton cycle are represented in Fig. 4.10.

A close cycle consists of four main stages, as follows:

- *Stage 1 to 2: Air compression through adiabatic process*—fresh air enters the compressor in stage 1. Through stage 1 to 2, air is compressing and increasing temperature whereas entropy is constant and volume of the air is decreasing. At stage 2, high pressure and high temperature air is discharged from the compressor.
- *Stage 2 to 3: Air combustion through isobaric processes*—in this stage, compressed air enters the combustion chamber. Fuel and air are mixed and burned at constant pressure. After the combustion process, gas volumetric expansion occurs and at stage 3, there is a very high temperature gas.
- *Stage 3 to 4: Air expansion through adiabatic process*—at stage 3, working fluid enters the turbine to drive it. When hot gas runs through the turbine, volume is larger than before and gas pressure is decreasing, so the turbine is called an expander.
- *Stage 4 to 1: Air heat transfer through isobaric process*—in stage 4, turbine exhaust gas is not proper for stage 1, thus heat transfer to the surrounding is necessary. After the heat transfer, there is a gas at atmospheric pressure as well as appropriate volume and temperature, which is ready to enter the compressor again.

The Brayton cycle network with respect to Fig. 4.10B is equal to area of a 1-2-3-4 close loop. Output efficiency of this cycle is an important parameter because it represents a reasonable operation or inappropriate performance of the gas turbine. The following equations are according to Fig. 4.10.

Added heat to cycle is equal to [8]:

$$Q_{in} = h_3 - h_2 = C_p(T_3 - T_2) \quad (4.1)$$

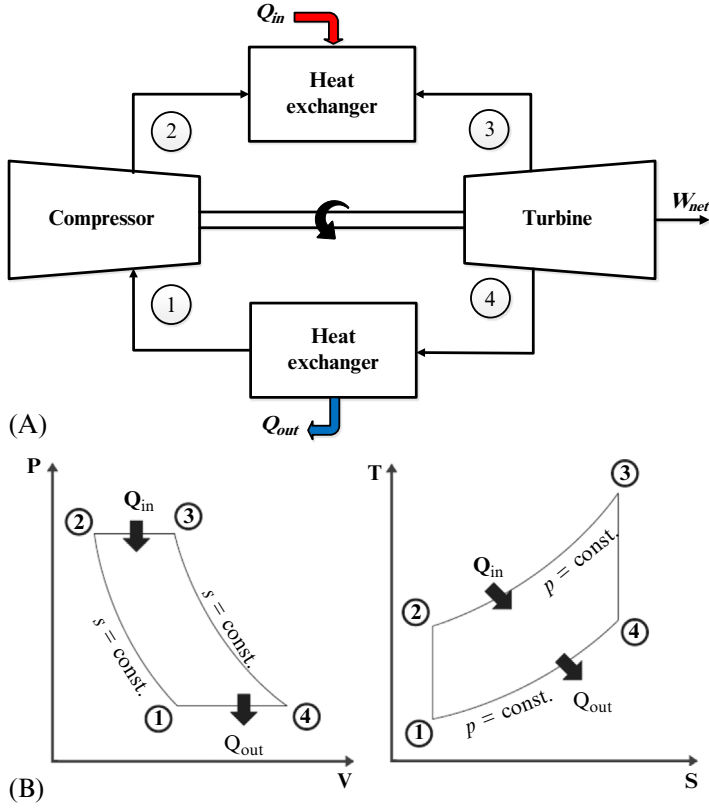


Fig. 4.10 Brayton cycle: (A) A close cycle gas turbine engine and (B) P-V and T-S diagram.

where C_p is heat capacity in constant pressure, T_2 and T_3 are temperatures in stages 2 and 3, respectively, and h is enthalpy. Heat rejection from turbine is as follows:

$$Q_{out} = h_4 - h_1 = C_p(T_4 - T_1) \quad (4.2)$$

From Eqs. (4.1) (4.2), net work is

$$W_{net} = Q_{in} - Q_{out} = C_p[(T_3 - T_2) - (T_4 - T_1)] \quad (4.3)$$

Brayton cycle efficiency is equal to the ratio of net work to added heat. Efficiency is as follows:

$$\eta = \frac{W_{net}}{Q_{in}} = \frac{C_p[(T_3 - T_2) - (T_4 - T_1)]}{C_p[(T_3 - T_2)]} = 1 - \frac{T_4 - T_1}{T_3 - T_2} \quad (4.4)$$

The isentropic process is an idealized thermodynamic process in which there is no heat transfer from the system to its surrounding. In the isentropic process, the following equation is true.

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{(\gamma-1/\gamma)} = \left(\frac{V_1}{V_2} \right)^{(\gamma-1)}, \quad \gamma = \frac{C_p}{C_v} = \frac{c_p}{c_v} \quad (4.5)$$

where γ is heat capacity ratio, C is heat capacity, and c is specific heat capacity that index p and v are related to constant pressure and constant volume, respectively. Similarly,

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4} \right)^{(\gamma-1/\gamma)} \quad (4.6)$$

According to Fig. 4.3, because $P_2 = P_3$ and $P_1 = P_4$, thus

$$\frac{T_2}{T_1} = \frac{T_3}{T_4} = \left(\frac{P_2}{P_1} \right)^{(\gamma-1/\gamma)} \quad (4.7)$$

$$\frac{T_4 - T_1}{T_3 - T_2} = \frac{T_1}{T_2} = \frac{T_3}{T_4} \quad (4.8)$$

From Eqs. (4.8) (4.4), efficiency is as follows:

$$\eta = 1 - \frac{T_1}{T_2} = 1 - \frac{T_3}{T_4} \quad (4.9)$$

Pressure ratio is called r_p and Eq. (4.10) is rewritten Eq. (4.7).

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{(\gamma-1/\gamma)} = (r_p)^{(\gamma-1/\gamma)} \quad (4.10)$$

Finally, the efficiency of the Brayton cycle is as follows:

$$\eta = 1 - \frac{1}{(r_p)^{\left(\frac{\gamma-1}{\gamma} \right)}} = 1 - (r_p)^{\left(\frac{1-\gamma}{\gamma} \right)} \quad (4.11)$$

Note, all mentioned equations are based on ignorance of kinetic energy and potential energy changes. According to Eq. (4.11), efficiency depends on temperature and pressure ratio but maximum and minimum temperatures are limited. Maximum temperature is limited by materials because turbine blades cannot withstand high temperatures and minimum temperature is defined by the temperature of inlet air to the engine.

Example 4.1

A gas turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 6. The gas temperature is 350°K at the compressor inlet and 1500°K at the turbine inlet. Utilizing the air-standard assumptions, determine (a) the gas temperature at the exits of the compressor and the turbine, (b) the back work ratio, and (c) the thermal efficiency.

Solution

The T-S diagram of the ideal Brayton cycle described is shown in Fig. 4.11. Note that the components involved in the Brayton cycle are steady flow devices.

- (a) The air temperatures at the compressor and turbine exits are determined from isentropic relations in Appendix B:

$$\text{Process 1-2: } T_1 = 350\text{K} \rightarrow h_1 = 350.49\text{kJ/kg}, P_{r1} = 2.379$$

$$(\text{At compressor exit}) P_{r2} = \frac{P_2}{P_1} P_{r1} = 6 \times 2.379 = 14.274 \rightarrow T_2 = 580\text{K},$$

$$h_2 = 586.04\text{kJ/kg}$$

$$\text{Process 3-4: } T_3 = 1500\text{K} \rightarrow h_3 = 1635.97\text{kJ/kg}, P_{r3} = 601.9$$

$$(\text{At turbine exit}) P_{r4} = \frac{P_4}{P_3} P_{r3} = \left(\frac{1}{6}\right) \times 601.9 = 100.32 \rightarrow T_4 = 970\text{K},$$

$$h_4 = 1011.9\text{kJ/kg}$$

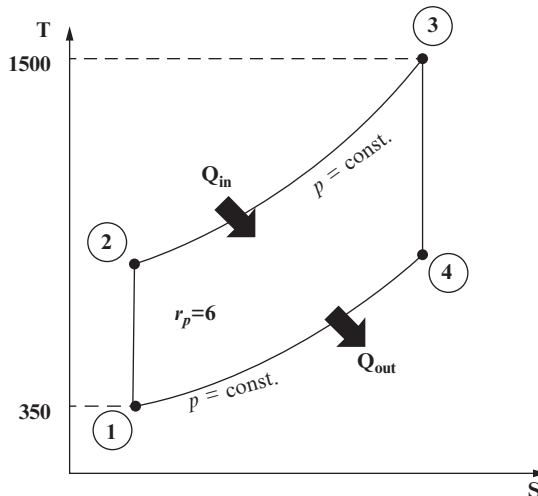


Fig. 4.11 Brayton cycle T-S diagram of Example 4.1.

- (b) To find the back work ratio, we need to find the work input to the compressor and the work output of the turbine:

$$\begin{cases} w_{comp,in} = h_2 - h_1 = 586.04 - 350.49 = 235.55 \text{ kJ/kg} \\ w_{turb,out} = h_3 - h_4 = 1635.97 - 1011.9 = 624.07 \text{ kJ/kg} \end{cases} \rightarrow r_{bw} = \frac{w_{comp,in}}{w_{turb,out}} = 0.377$$

That is, 33% of the turbine work output is used just to drive the compressor.

- (c) The thermal efficiency of the cycle is the ratio of the net power output to the total heat input:

$$\begin{cases} Q_{in} = h_3 - h_2 = 1635.97 - 586.04 = 1049.93 \\ w_{net} = w_{out} - w_{in} = 624.04 - 235.55 = 388.49 \end{cases} \rightarrow \eta_{th} = \frac{w_{net}}{Q_{in}} = 0.37 \text{ or } 37\%$$

Example 4.2

In a Brayton cycle-based power plant, the air at the inlet is at 27°C, 0.1 MPa. The pressure ratio is 6.25 and the maximum temperature is 800°C. Find (a) the compressor work per kilograms of air, (b) the turbine work per kilograms of air, (c) the heat supplied per kilograms of air, and (d) the cycle efficiency. (Consider $\gamma = 1.4$, which is the specific heat ratio value of air at room temperature)

Solution

Based on mentioned values: $T_1 = 27^\circ\text{C} = 300 \text{ K}$, $P_1 = 100 \text{ kPa}$, $r_p = 6.25$, $T_3 = 800^\circ\text{C} = 1073 \text{ K}$, and thus

- (a) Process 1–2:

$$\begin{aligned} \frac{T_2}{T_1} &= r_p^{\left(\frac{\gamma-1}{\gamma}\right)} = 6.25^{\left(\frac{1.4-1}{1.4}\right)} = 1.689 \rightarrow T_2 = 506.69 \text{ K}, W_{comp} \\ &= c_p(T_2 - T_1) = 1.005(506.69 - 300) = 207.72 \text{ kJ/kg} \end{aligned}$$

- (b) Process 3–4:

$$\begin{aligned} \frac{T_3}{T_4} &= r_p^{\left(\frac{\gamma-1}{\gamma}\right)} = 6.25^{\left(\frac{1.4-1}{1.4}\right)} = 1.689 \rightarrow T_4 = 635.29 \text{ K}, W_{turb} \\ &= c_p(T_3 - T_4) = 1.005(1073 - 635.29) = 439.89 \text{ kJ/kg} \end{aligned}$$

- (c) Heat input,

$$Q_{in} = c_p(T_3 - T_2) = 1.005(1073 - 506.69) = 569.14 \text{ kJ/kg}$$

(d) Cycle efficiency,

$$\eta_{th} = \frac{(W_{turb} - W_{comp})}{Q_{in}} = \frac{439.89 - 207.72}{569.14} = 0.408 \text{ or } 40.8\%$$

There are three general methods for improving Brayton cycle efficiency: regeneration, reheat, and intercooling. Fig. 4.12 shows a Brayton cycle that simultaneously uses reheating, intercooling, and regeneration methods.

T-S and P-V diagrams for improved Brayton cycle are given in Fig. 4.13. According to these diagrams, surface increasing causes more net work and higher efficiency.

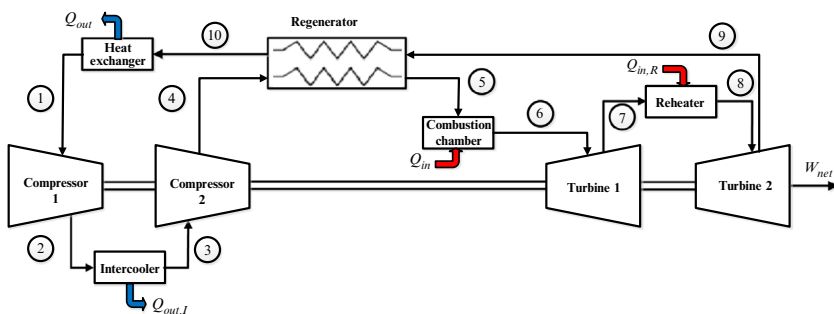


Fig. 4.12 Improved Brayton cycle.

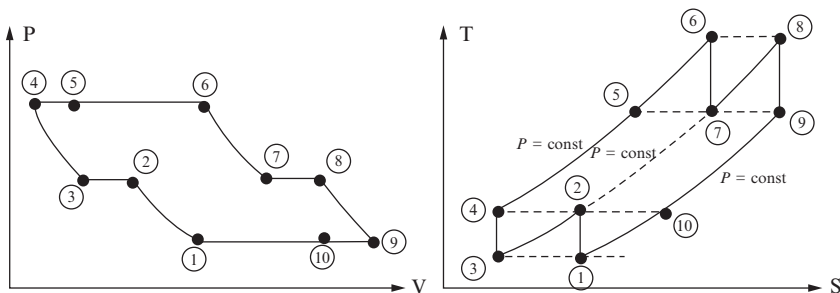


Fig. 4.13 Diagrams of improved Brayton cycle.

Example 4.3

Solve Example 4.2 if a regenerator of 80% effectiveness is added to the plant.

Solution

With respect to Fig. 4.14,

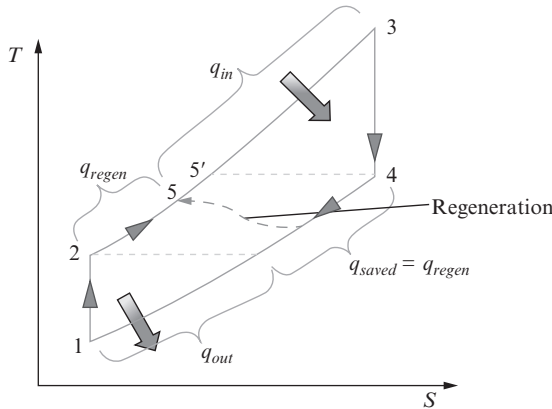


Fig. 4.14 T-S diagram with a regenerator of Example 4.3.

$$\varepsilon = \frac{T_5 - T_2}{T_4 - T_2} = 0.8 \rightarrow 0.8 = \frac{T_5 - 506.69}{635.29 - 506.69} \rightarrow T_5 = 603.14\text{K}$$

and T_4 , W_{comp} , W_{turb} remain unchanged.

The new input heat is equal to:

$$Q_{in} = c_p(T_3 - T_5) = 472.2\text{kJ/kg}$$

so,

$$\eta_{th} = \frac{(W_{turb} - W_{comp})}{Q_{in}} = \frac{439.89 - 207.72}{472.2} = 0.492\% \text{ or } 49.2\%$$

4.5 MT COMPONENTS

MTG basic components are a compressor for creating appropriate pressure and a combustor and turbine that are coupled to a generator. An induction generator or synchronous generator is used in MTGs but a permanent magnet synchronous generator (PMSG) is superior. The field winding is replaced by a permanent magnet in PMSGs, which cause lower power losses, higher mechanical consistency, structure simplicity, and lower maintenance; so PMSGs are mostly used in MTGs. To get higher efficiency, a recuperator, which is a heat exchanger, is added to other components. In a single-shaft design, all components are mounted on the same shaft. Fig. 4.15 shows a MTG configuration in detail. This section provides a brief description of MTG components.

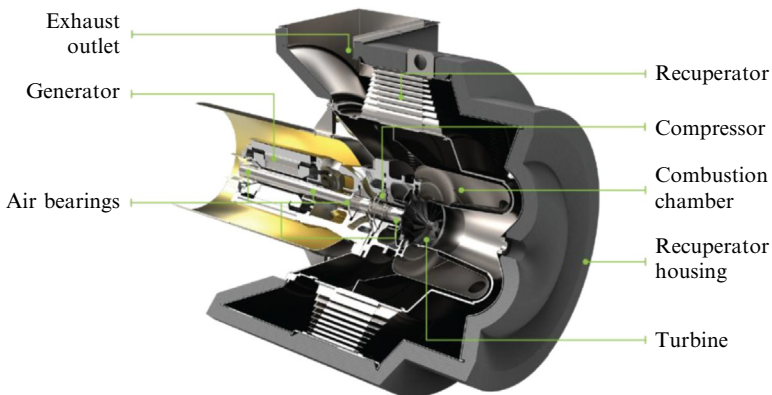


Fig. 4.15 Cross section of a microturbine generation system [9].

4.5.1 Compressor

MTs require high-pressure gases for appropriate combustion, thus a compressor is used. A gas compressor is a mechanical device that increases the pressure of a gas by reducing its volume, and an air compressor is a specific type of gas compressor. An air compressor is used to increase the pressure of inlet air at ambient temperature by using some external energy. The compressors are divided into dynamic and positive displacement, which differ in energy transfer. In a dynamic compressor, energy transfer is done continuously but in a positive displacement type transfer is periodic.

There are two main types of positive displacement compressors: reciprocating and rotary. In the reciprocating compressor, air volume is drawn into a cylinder and compression is done in it. Airflow is controlled by a cylinder valves piston and this valve acts as a check valve. In a rotary compressor, gases volume is decreased by lobes, screws, and vanes.

A dynamic type of compressor is used in gas turbines. Dynamic type includes a radial flow compressor and an axial flow compressor. In a radial or centrifugal compressor, air is compressed through an impeller or rotating disk by use of centrifugal force. After the impeller, flow enters a diffuser and static pressure is increased by reducing velocity. In other words, a centrifugal compressor accelerates the gasses velocity (increase in kinetic energy), which is then converted into pressure as the airflow leaves the volute and enters the discharge pipe. A centrifugal compressor delivers higher flow rates than a positive displacement compressor. An axial compressor is composed of a rotor that has rows of fans as blades where rotating blades are attached to a shaft to push air over stationary blades called stators. The stator blades

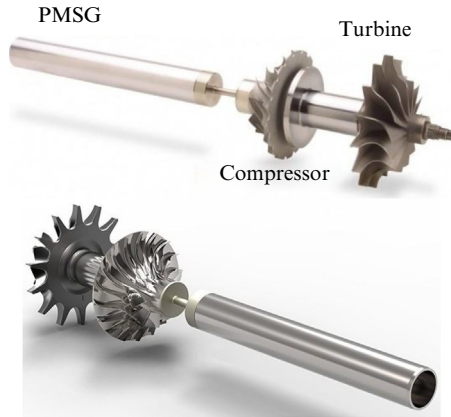


Fig. 4.16 Single-shaft microturbine components [9].

are attached to the casing. Gas or working fluid flows parallel to the rotating shaft and rotating blades increase air velocity, which is down by the stationary blades. The decreasing air velocity converts kinetic energy to pressure. Centrifugal type compressors are simpler and lower cost than axial type and are favorable for MTs. Most MTGs are single-stage centrifugal compressors, and the axial flow compressor is more suitable for a heavy gas turbine. Fig. 4.16 shows a centrifugal compressor in MTs that is coupled with a turbine.

Compressor efficiency has a direct influence on MTG overall efficiency because output mechanical power of the turbine is consumed by a compressor that are coupled together in a same shaft. Recently, compressor manufacturers have used advanced computational fluid dynamics to obtain higher efficiency.

Compressor ratio is defined as the ratio of air volume before compression to air volume after compression, and pressure ratio between 3:1 or 4:1 is best suitable for a single-stage centrifugal compressor. Fig. 4.17 shows electrical efficiency of MT as a function of compressor pressure ratio for various turbine inlet temperatures and optimum performance exists in the pressure ratio range of 2.5–4.5.

Specific power of the MT as a function of compressor pressure ratio for the same range of firing temperatures as last figure is shown in Fig. 4.18.

Greater specific power can be obtained with higher pressure ratios but higher pressure ratio is depending on higher rotational speed. Rotational speed is limited by turbine blade materials, so current pressure ratio is appropriate in cost and operation.

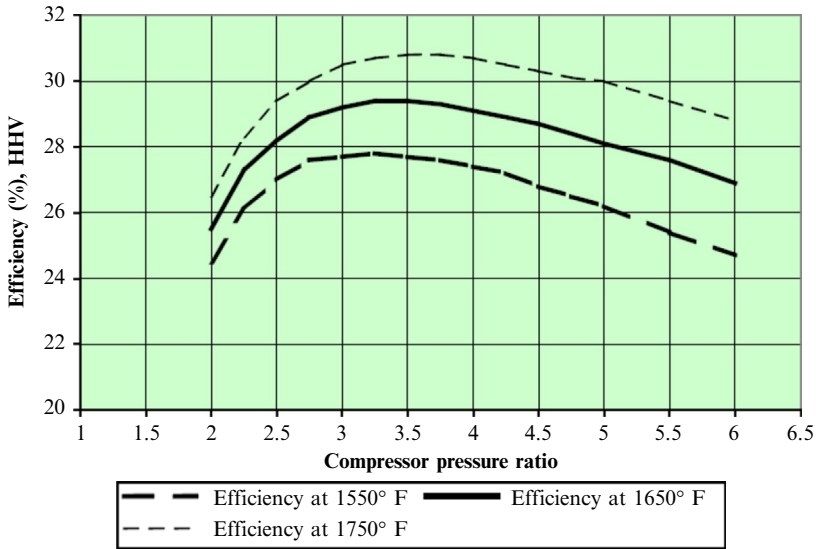


Fig. 4.17 Microturbine efficiency as a function of compressor pressure ratio and turbine inlet temperature [10].

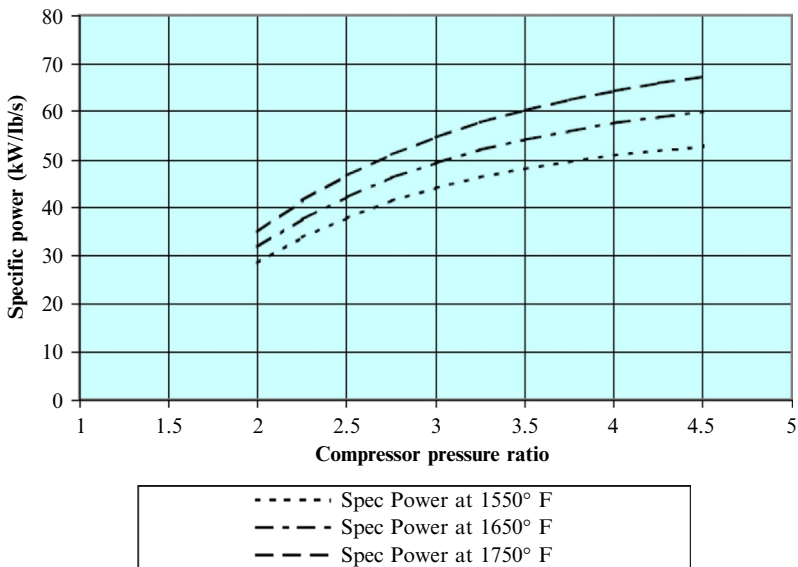


Fig. 4.18 Microturbine specific power as a function of compressor pressure ratio and turbine inlet temperature [10].

4.5.2 Turbine

The turbine is the heart of MTs because it plays the main role in electric power production and in single-shaft design, where a compressor and a generator are rotating with the turbine. A turbine is a rotary mechanical device, which is used to convert the energy of a working fluid to useful work. The working fluid contains potential energy and kinetic energy and the turbine extracts energy by several physical principles.

The main factors in choosing turbines include manufacturing costs, cycle flow path, leakage problems, and stability of shaft. Axial and radial are two types of turbine that are different in air path flow. In the radial turbine the flow of working fluid is radial to the shaft. The axial types are used in heavy gas turbines but radial types are more efficient in micro gas turbines. The advantages of the use of radial turbines in MTs are higher single-stage ratio, small volume, simple structure, and higher efficiency. Fig. 4.16 shows a radial turbine that is coupled to a compressor.

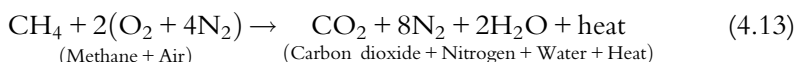
4.5.3 Combustor

Combustion means the act of burning, and the combustion process is a high-temperature chemical reaction between the fuel and oxygen in which heat is released.

A combustion chamber is a part of an internal combustion engine in which fuel and compressed air are mixed and burned; the resulting high-temperature exhaust gas is used to drive the turbine. The combustion chamber in a gas turbine is called a combustor, and a MT combustor has continuous operation unlike in a reciprocating engine. The combustion equation of the natural gas is given by [11]



Because air contains 78% nitrogen and 22% oxygen, there are four molecules of nitrogen for every molecule of oxygen in air. So, the complete combustion reaction of methane can be written as follows [11]:



The combustor's purpose is to produce enough energy for the system. The main factors in a combustor's design include complete combustion, lower pressure losses, smaller size, and lower environmental emissions. There are three main types of combustor: annular type, can type, and can-annular type, which are briefly described below.

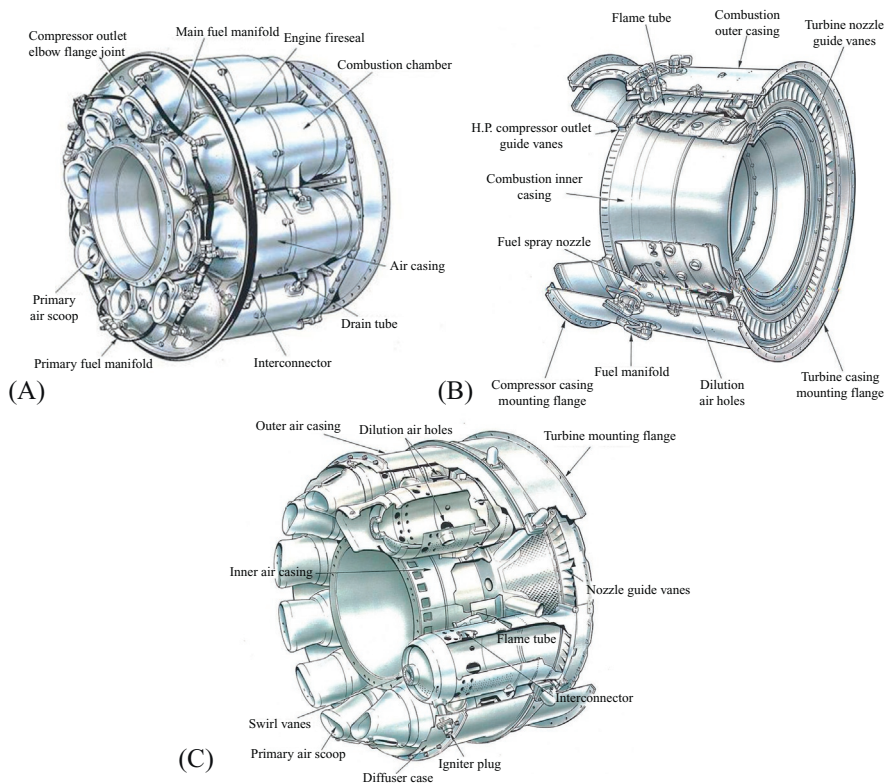


Fig. 4.19 Types of gas turbine combustors: (A) can, (B) annular, and (C) can-annular [11].

- *Can type*: This type is the earliest one and consists of separate cans. Each “can” has its own fuel injector, liner, interconnectors, and casing. Fig. 4.19A shows a can type of combustor.
- *Annular type*: This type is a modern combustor that is shown in Fig. 4.19B. Annular combustors have separate combustion zones, a continuous liner, and casing in a ring (the annulus). Annular designs have more efficient combustion in which nearly all of the fuel is burned completely.
- *Can-annular type*: This type is combination of the two abovementioned types, with discrete combustion zones contained in separate liners with their own fuel injectors and all the combustion zones share a common air casing. Can-annular designs have less-efficient combustion but their modular design is easy to repair or replace. Fig. 4.19C shows can-annular type combustor.

The annular type of combustor is usually used in MTs and integrated design. Advantages of the annular type include shorter size, more uniform

combustion, less surface area, permitting better mixing of the fuel and air as well as simple structure. With respect to recent discussions, features of annular combustor are suitable for MT.

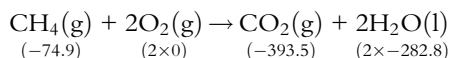
Example 4.4

With respect to Table 4.1, find the higher heating value (HHV) of methane CH_4 in kJ/mol and kJ/kg when it is oxidized to CO_2 and liquid H_2O [12].

Solution

First, HHV and lower heating value (LHV) are defined. The HHV, also known as the gross heat of combustion, includes latent heat, while the LHV, or net heat of combustion, does not.

The reaction is written below, and beneath it are enthalpies taken from Table 4.1. Notice that the equation must be balanced so that we know how many moles of each constituent are involved.



Notice that we have used the enthalpy of liquid water to find the HHV. The difference between the total enthalpy of the reaction products and the reactants is

$$\begin{aligned} \Delta H &= [(-393.5) + 2 \times (-282.8)] - [(-74.9) + 2 \times 0] \\ &= -890.2 \text{ kJ/mol of } \text{CH}_4 \end{aligned}$$

Because the result is negative, heat is released during combustion; that is, it is exothermic. The HHV is the absolute value of ΔH , which is 890.2 kJ/mol . As there are $12.011 + 4 \times 1.008 = 16.043 \text{ g/mol}$ of CH_4 , the HHV can also be written as

$$\text{HHV} = \frac{890.2 \text{ kJ/mol}}{16.043 \text{ g/mol}} \times 1000 \text{ g/kg} = 55,490 \text{ kJ/kg}$$

Table 4.1 Enthalpy of formation H° at 1 atm, 25°C for selected substances

Substance	State	H° (kJ/mol)
H	Gas	217.9
O	Gas	247.5
O_2	Gas	0
H_2O	Liquid	-285.8
H_2O	Gas	-241.8
CH_4	Gas	-74.9
CO_2	Gas	-393.5

4.5.4 Recuperator

Heat exchangers are used to transfer heat from the turbine exhaust flow to the cycle of power generation. Recuperator and regenerator are two main types of heat exchanger. A recuperator operates based on direct heat exchange and there are separate flow paths for each fluid, which flow simultaneously through the heat exchanger. A regenerator is on the basis of an intermediate storage in which there is one flow path that hot and cold fluids alternately pass through. In other words, a recuperator is a heat exchanger for transferring heat from the hot turbine discharges to the colder compressed air.

MTs are required to have a recuperator to achieve desirable thermodynamic efficiency. Adding heat to compressed air causes a reduction in the amount of fuel required to raise the temperature of gases used by the turbine, so a recuperator in MT is necessary for fuel and cost reduction and improving efficiency. However, the recuperator causes additional pressure losses leading to reduction of overall efficiency. Therefore, the recuperator has positive and negative effects on output efficiency and then balancing is required. There are connections between the recuperator and the compressor discharge, the expansion turbine discharge, the combustor inlet, and the system exhaust. Thus, design of recuperator configuration is a great challenge because it is required to lower pressure losses, lower manufacturing costs, and increase reliability in high temperatures. Main factors of the recuperator include high effectiveness, low pressure losses, minimum volume and weight, high reliability and durability, and low cost. The recuperator impact on efficiency of the MT is shown in Fig. 4.20. The inclusion of a high effectiveness recuperator increases the electric efficiency of a MT [10].

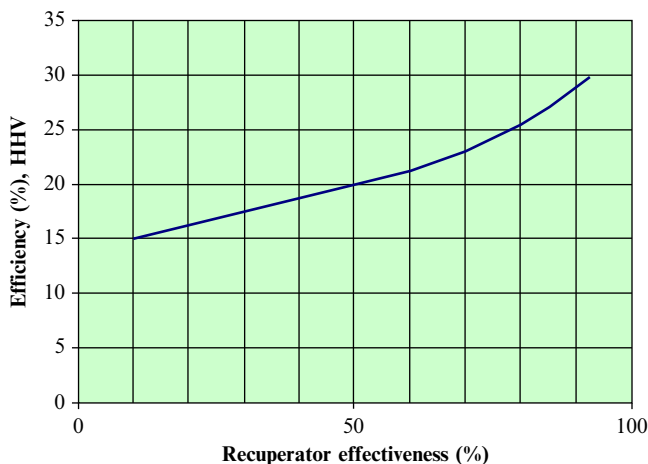


Fig. 4.20 Microturbine efficiency as a function of recuperator effectiveness [10].

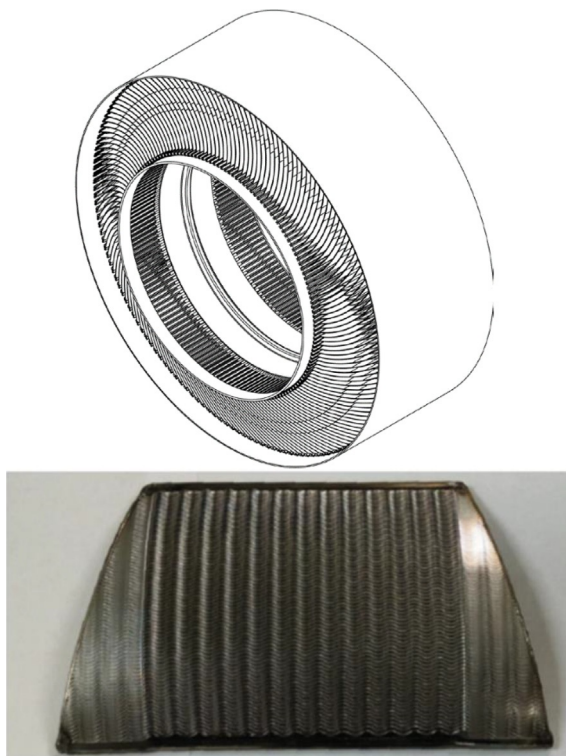


Fig. 4.21 Annular primary surface stainless steel recuperator.

Recuperators are divided into annular wraparound and rear-mounted cube types. The advantages of the annular type are compact design, good aerodynamic gas flow path, low pressure drop, no need for external ducts, and lower acoustic signature. The advantages of cube shape are simplicity of hot gas bypass for cogeneration and an external combustor for a variety of dirty fuels. Fig. 4.21 shows a recuperator for small gas turbine.

Recuperator design is an important issue for limiting deformation and thermal fatigue. Stainless steel is the typical material for a recuperator and it does provide long life with high efficiency. Some materials used in the structure of a recuperator are 300 series stainless steel, Inconel 625, Inconel 803, Haynes 120, Haynes 214, and PM2000.

4.5.5 Permanent Magnet Synchronous Generator

The term PMSG refers to two facts about its configuration. Permanent magnet means a generator without excitation coils (DC field winding),

which are replaced by a permanent magnet. Synchronous means rotor and magnetic field, which are rotating with the same speed and frequency of induced voltage in the stator directly proportional to revolutions per minute (rpm). Today, PMSGs are widely used in steam turbines, gas turbines, hydro turbines, and wind turbine for converting mechanical to electrical output power for users. Generally, in a MT generator a PMSG is used instead of an induction generator because of following advantages [13]:

- No significant losses generated in the rotor
- Lower maintenance cost
- Design flexibility
- Very high torque is achieved at low speed
- Eliminates the need for separate excitation system
- Higher power density and higher reliability

However, the disadvantages of PMSG are

- Higher initial cost
- Flexibility loss of field flux control
- Possible demagnetization and thermal stress

Two general types of PMSGs are salient pole or nonsalient pole on average 70% efficiency. Construction materials typically are neodymium-iron-boron (NdBFe) or samarium-cobalt magnets, which are very suitable for such high-speed electrical machines. Fig. 4.9 shows a permanent magnet that is used in MTs, and a cross section of PMSG is shown in Fig. 4.22.

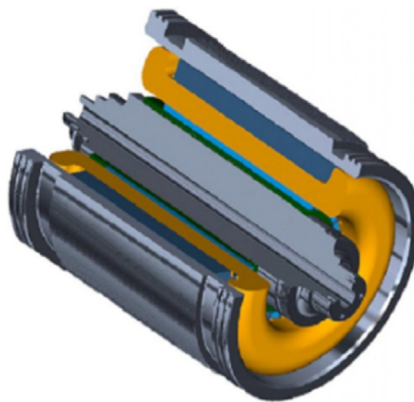


Fig. 4.22 Cross section of a permanent magnet synchronous generator.

4.6 MT APPLICATIONS

Recently, power sources with high-efficiency, reliability, and low pollutant emission level have been considered. MTs have several factors that are acceptable and appropriate for use in various applications. In the following sections, the most important and popular applications of MT are discussed.

4.6.1 Peak Shaving

One of the most important applications of MTs is using them in peak load periods. In peak demand, MTs are used to reduce costs and improve overall efficiency as well as reduce investments in bulk transmission. In general, peaking power plants are run only when there is peak demand for electricity. The MT start-up time is about 1–2 min. Moreover, appropriate output power allows using it in periods of high demand for peak shaving, which reduces overall costs. Peak shaving in combination with emergency power or standby power is economically advantageous.

4.6.2 Combined Heat and Power

Cogeneration or CHP is using wasted heat to generate electricity and useful heat (or cooling (CCHP)) at the same time. During normal operation, MTs produce significant amounts of high-temperature exhaust gases that with proper management and use of them, system overall efficiency reaches up to 80%. Wasted heat is used in forms of warm water, steam, direct use like heating or drying, driving thermally activated equipment, as well as chilling water. Fig. 4.23 shows MTG-based CHP.

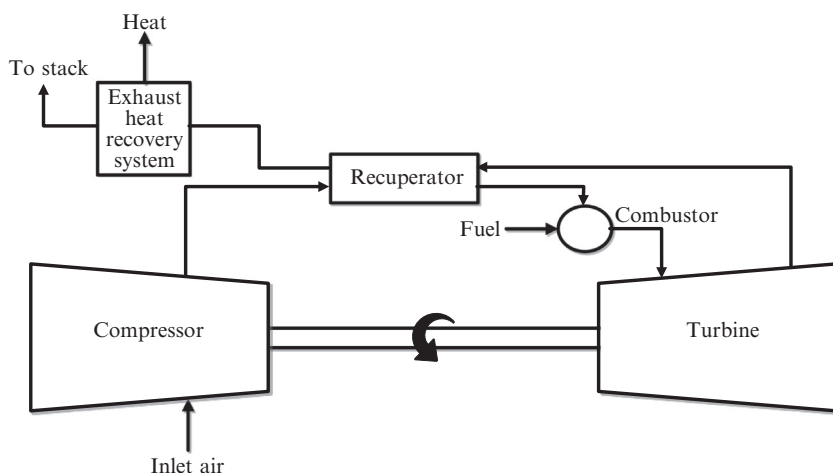


Fig. 4.23 Microturbine-based combined heat and power system.

Generally, the advantages of CHPs, which have created a global trend toward their use, include competitive capital investment, high efficiency, high power quality, environmental friendliness (low pollutant emissions level of NO_x , CO, and CO_2), and operational flexibility. These CHP advantages are in addition to MT benefits such as simple installation, light weight, and low maintenance; hence, this combination is suitable. Typically the range of MT, which is used in CHP, is around 100 kW and they are known as micro CHPs.

Example 4.5

The Elliott TA 100A at its full 105 kW output burns 1.24×10^6 Btu/h of natural gas. Its waste heat is used to supplement a boiler used for water and space heating in an apartment house. The design calls for water from the boiler at 120°F to be heated to 140°F and returned to the boiler. The system operates in this mode for 8000 h per year [12].

- If 47% of the fuel energy is transferred to the boiler water, what should the water flow rate be?
- If the boiler is 75% efficient, and it is fueled with natural gas costing \$6 per million Btu, how much money will the MT save in displaced boiler fuel?
- If utility electricity costs \$0.08 kWh, how much will the MT save in avoided utility electricity?
- If operation and maintenance is \$1500 year, what is the net annual savings for the MTe?
- If the MT costs \$220,000, what is the ratio of annual savings to initial investment (called the initial rate of return)?

Solution

- The heat Q required to raise a substance with specific heat c and mass flow rate $\dot{m}$ by a temperature difference of ΔT is:

$$Q = \dot{m}c\Delta T$$

Because it takes 1 Btu to raise 1 lb of water by 1°F , and 1 gal of water weighs 8.34 lb, we can write

$$\begin{aligned} \text{Water flow rate } \dot{m} &= \frac{0.47 \times 1.24 \times 10^6 \text{ Btu/h}}{1 \text{ Btu/lb}^\circ\text{F} \times 20^\circ\text{F} \times 8.34 \text{ lb/gal} \times 60 \text{ min/h}} \\ &= 58 \text{ gpm} \end{aligned}$$

- The fuel displaced by not using the 75% efficient boiler is worth

$$\begin{aligned} \text{Fuelsaving} &= \frac{0.47 \times 1.24 \times 10^6 \text{ Btu/h}}{0.75} \times \frac{\$6.00}{10^6 \text{ Btu}} \times 8000 \text{ h/year} \\ &= \$37,300 \text{ year} \end{aligned}$$

(c) The utility electricity savings is

$$\begin{aligned}\text{Electric utility savings} &= 105 \text{ kW} \times 8000 \text{ h/year} \times \$0.08/\text{kWh} \\ &= \$67,200/\text{year}\end{aligned}$$

(d) The cost of fuel for the MT is

$$\begin{aligned}\text{Microturbine fuel cost} &= 1.24 \times 10^6 \text{ Btu/h} \times \frac{\$6.00}{10^6 \text{ Btu}} \times 8000 \text{ h/year} \\ &= \$59,520/\text{year}\end{aligned}$$

So the net annual savings of the MT, including \$1500 year in operation and maintenance, is MT savings = (\$37,300 + \$67,200) - \$59,520 - \$1500 = \$43,480 year

(e) The initial rate of return on this investment would be:

$$\begin{aligned}\text{Initial rate of return} &= \text{Annual saving} / \text{Initial investment} = \\ &= \frac{\$43,480/\text{year}}{\$220,000} = 0.198 = 19.8\%/\text{year}\end{aligned}$$

4.6.3 Backup Generation

In this application, MTs are used as standby power or uninterruptible power supply (UPS). A standby generator is used in event of an outage, as a backup electrical system that operates automatically. Similarly, an UPS is an electrical device, which produces emergency power to load. The MTs are an ideal choice to use as an UPS and standby power because of their low initial cost, high level of reliability, and low maintenance requirement. In this situation, grid reliability for supplying of sensitive loads is increased due to the high level of support.

4.6.4 Resource Recovery

Selective extraction of disposed materials for a specific next use to extract maximum benefits from products is called resource recovery. In this application, MTs burn waste gasses that are flared or released directly to the atmosphere. MTs can convert unprocessed gas that contain up to 7% H₂S to electricity and this reduces fuel consumption and costs. In addition to its fuel flexibility, the MT was introduced as a more efficient and most economical power source.

4.6.5 Transportation Application

Recently, the use of MTs has increased in hybrid electric vehicles (HEVs) because of their high efficiency and low pollutant emissions. MT application

in transportation includes transit busses, heavy trucks, HEVs, and super cars. The MTs enabled increased mileage without refueling (reduction of fuel consumption reaches up to 40%). The MTGs with low levels of emissions and noise and high efficiency are appropriate cases for a transportation system.

4.6.6 Base Load

MTs are used as base load to increase power quality and reliability. Base load power sources can consistently generate the electrical power needed to satisfy minimum demand (minimum amount of electric power delivered or required over a given period at a constant rate). Generally, continuous operation of the MTs in rated capacity cause its efficiency to improve.

4.6.7 Hybrid System

Recently, research has focused on the combination of two or more DGs that are known as hybrid systems. To improve the reliability of renewable DGs, a MT is an appropriate choice and the combination of a fuel cell and a MT is a desired and popular hybrid system with overall efficiency that reaches up to 50%. Advantages of hybrid system include reduction of investment, lower pollutant emission, higher reliability, fuel consumption reduction, and higher efficiency.

4.6.8 Remote Power

In locations where network construction is impossible or it is not economically beneficial, MTs are useful. Typically, cost of grid construction is more than a DG. The MTs fuel flexibility is a determining factor for use in remote applications, which causes significant reduction of investment costs.

Mentioned applications are most important in MTGs but there are other applications, which are not included in this section. Generally, the main purpose is to achieve high efficiency, reliability, and economic power source.

4.7 TYPES OF MT

The main parts of a MT are the compressor, combustor, and turbine. As mentioned earlier, the compressor is located before the turbine and the combustion chamber is between them. Inlet air at ambient temperature is compressed by the compressor and then compressed air is mixed with fuels in the combustor. High temperature and pressure gases drive the turbine and

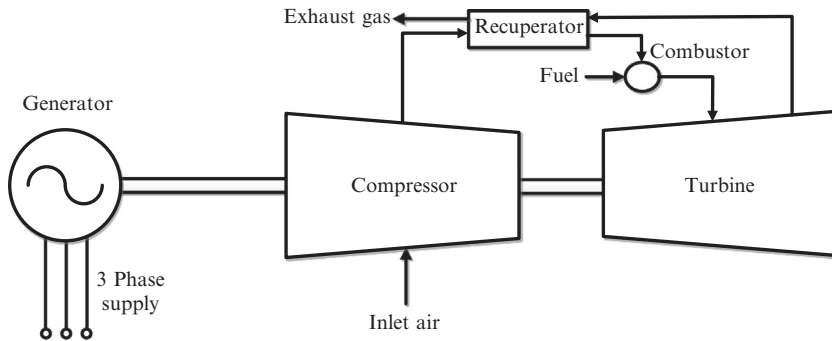


Fig. 4.24 Single-shaft microturbine generation system.

produce mechanical power. A compressor and a generator are required turbine output mechanical power for rotating. This cycle is true in gas turbines and its subsets.

The shaft structure defines important characteristics of the MT that eventually influence the required power electronics and control system. There are two types of MT design: single-shaft design and split-shaft or two-shaft design. Single-shaft design of a MT is represented in Fig. 4.24.

In this model, all components are mounted on same shaft and generator and compressor operation are dependent on the turbine. In the shingle-shaft design, the speed of the shaft is about 50,000–120,000 rpm, which causes high-frequency output voltages. High-frequency voltage is needed to power electronic interfaces for converting to the desired frequency for customers, which is discussed later.

Split-shaft design is not a required power electronic interface and Fig. 4.25 shows this type of MT. In the split-shaft design, there is a turbine for driving the compressor and another turbine for driving generators, which are coupled together through a gearbox. Speed of the shaft is about 3600 rpm and the output voltage frequency is 60 Hz (or 50 Hz) without using any converters. More moving parts cause a negative effect on efficiency and maintenance costs.

In general, single-shaft MTs are more popular because of higher flexibility, reliability, and lack of required lubrication.

4.8 TYPES OF POWER ELECTRONIC INTERFACE TOPOLOGIES

Power electronic interfaces are required in single-shaft MT designs because MT output power is high frequency, which is not appropriate for customers.

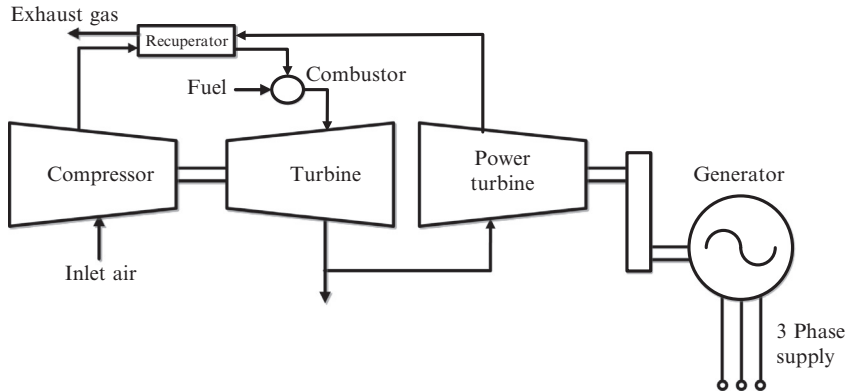


Fig. 4.25 Split-shaft microturbine generation system.

There are a variety of methods for converting of the MTs output power to desired power that is based on AC/DC/AC or AC/AC structures and there is not absolute excellency for any of them.

In an AC/DC/AC design based on a DC-link capacitor, high-frequency voltage is converted to DC voltage and then back to the desired frequency through an inverter. A controlled rectifier or active rectifier and uncontrolled rectifier or passive rectifier are considered as types of rectification. A passive rectifier is applied by diodes and this structure is simple and low cost but the advantages of an active rectifier are more flexibility and providing MTs starting-up features. The disadvantages of DC-link converters are large physical dimensions, lower lifetime, heavy weight, and lower reliability. Fig. 4.26 shows two typical structures of DC-link converters. However, a high-frequency link converter (HFLC) is another method of AC/DC/AC, which is shown in Fig. 4.27. One advantage of the HFLC is robust isolation, but it is more expensive with higher power electronics parts and losses that lead to less use in industry. Generally, power electronic interfaces based on a passive rectifier and inverter combination are the most common method because of their potential merits.

Recently, attention to AC/AC interface topology has increased. Two typical types of this structure are cycloconverters and matrix converters, which are shown in Fig. 4.28. Input AC voltage with arbitrary amplitude and frequency is converted to the desired frequency. These converters are without a DC-link capacitor that causes higher lifetime and efficiency. Other advantages of AC/AC structure include lower thermal stress on semiconductor switches, higher reliability, compact size, bidirectional power flow, and controllable input power factor. A major drawback of

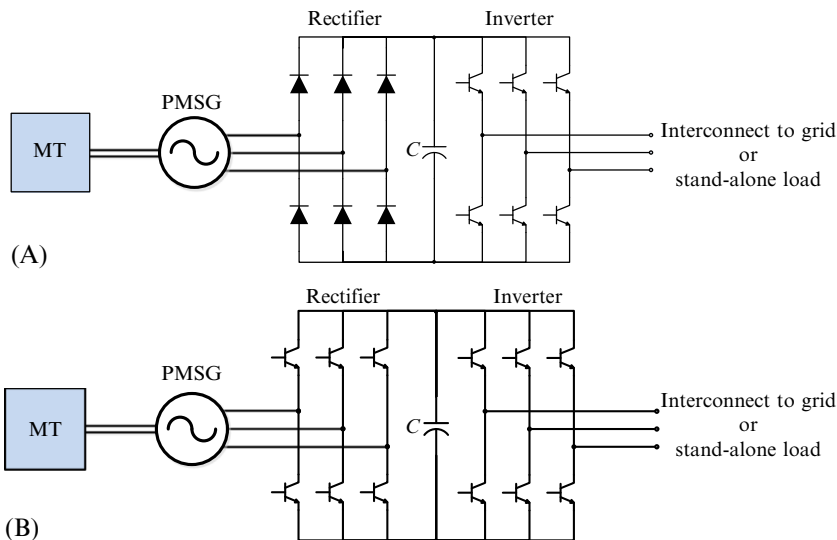


Fig. 4.26 Microturbine generation system power electronic interface: (A) passive rectifier and (B) active rectifier.

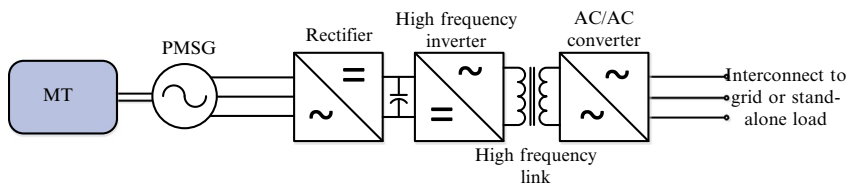


Fig. 4.27 Microturbine generation system with a high-frequency link power converter.

AC/AC converters is more switches lead to more power losses. Moreover, using of battery or other power sources and energy storage is not possible. These converters are sensitive to disturbance of input voltage and input fluctuation directly influences the output and vice versa. Control complexity and lower voltage transfer ratio are other disadvantages of these topologies.

None of the abovementioned structurea is superior. The simplest, least costly, and most common method for power electronic interface is the AC/DC/AC structure based on diode rectification, which is the simulation model of this chapter. In the following, converter simulation and dynamic model of MT with controls block diagrams will be considered.

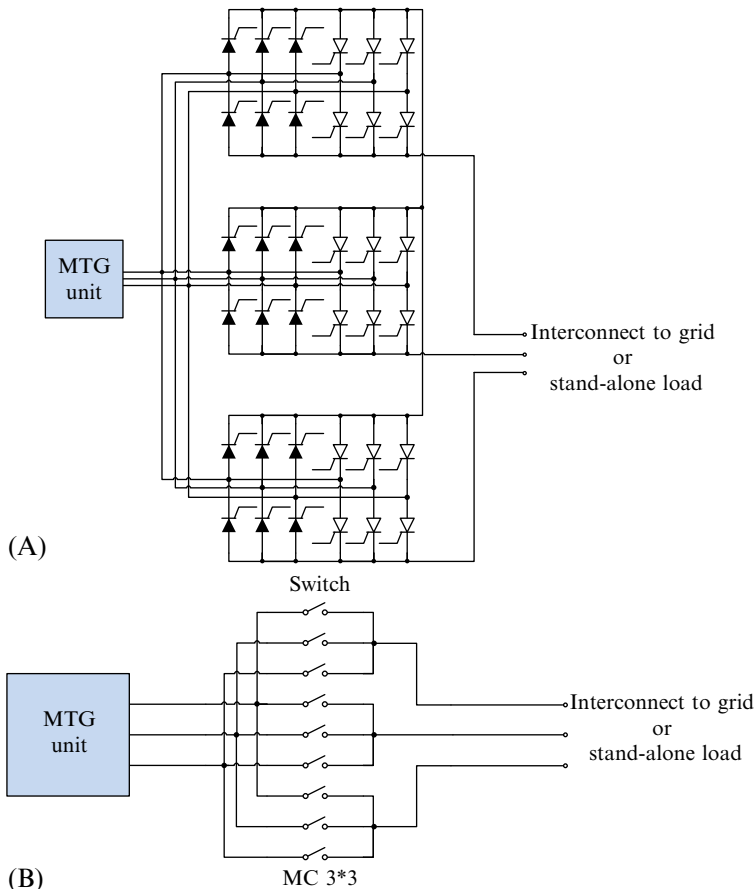


Fig. 4.28 Microturbine generation system power electronic interface: (A) cycloconverter and (B) matrix converter.

Example 4.6

A MT has a natural gas input of 13,700 Btu (LHV) per kWh of electricity generated. Find its LHV efficiency and its HHV efficiency [12] (Table 4.2).

Solution

Generally, the thermal efficiency of power plants is often expressed as a heat rate, which is the thermal input (Btu or kJ) required to deliver 1 kWh of electrical output (1 Btu/kWh = 1.055 kJ/kWh). The smaller the heat rate, the higher the efficiency. Heat rates are usually expressed in Btu/kWh, which results in the following relationship between it and thermal efficiency,

Table 4.2 Higher heating value and lower heating value for various fuels

Fuel	Higher heating value (HHV)		Lower heating value (LHV/HHV)		LHV/HHV
	Btu/lbm	kJ/kg	Btu/lbm	kJ/kg	
Methane	23,875	55,533	21,495	49,997	0.9003
Propane	21,669	50,402	19,937	46,373	0.9201
Natural gas	22,500	52,335	20,273	47,153	0.9010

$$\text{Heat rate (Btu/kWh)} = \frac{3412 \text{ Btu/kWh}}{\eta}$$

Using the LHV for fuel gives the LHV efficiency:

$$\text{Efficiency (LHV)} = \frac{3412 \text{ Btu/kWh}}{13,700 \text{ Btu/kWh}} = 0.2491 = 24.91\%$$

Using the LHV/HHV ratio of 0.9010 for natural gas in the above equation, gives the HHV efficiency for this turbine: $\text{Efficiency (HHV)} = 24.91\% \times 0.901 = 22.44\%$.

4.9 CONSTRUCTION AND OPERATION OF MTs

In this section, different aspects of the MT along with several diagrams are presented. The information is about overall efficiency, output power, and pollutant emissions, which are main factors in the construction and operation of MTs.

The ambient conditions at the inlet of the MT affect both power output and efficiency. At inlet air temperatures above 59°F (15°C), both the power and efficiency decrease. The power decreases due to the decreased air density with increasing temperature and the efficiency decreases because the compressor requires more power to compress higher-temperature air. Conversely, the power and efficiency increase when the inlet air temperature is below 59°F. Fig. 4.29 shows the variation in power and efficiency for a MT as a function of ambient temperature relative to the reference International Organization for Standards (ISO) condition of sea level and 59°F.

It should be noted that some manufacturers may place limitations on maximum power output below a certain ambient temperature due to maximum power limitations of the gearbox, generator, or power electronics, and this may modify the shape of the curves shown in Fig. 4.29 [10].

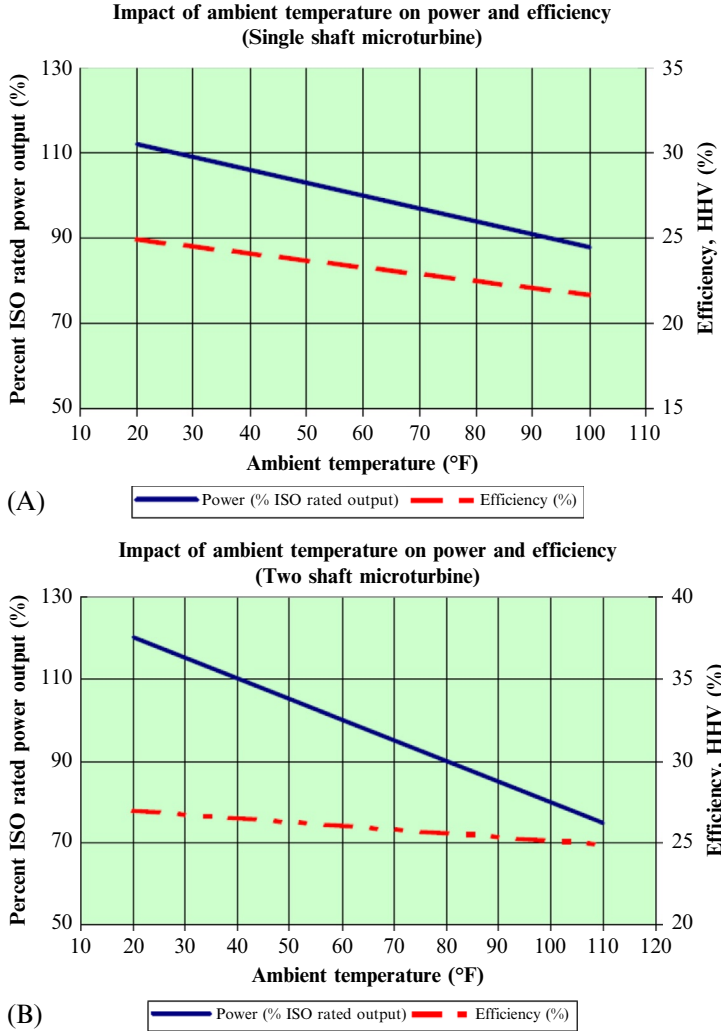


Fig. 4.29 Ambient temperature effects on microturbine performance: (A) single-shaft and (B) split-shaft [10].

Elevation has influence on the MT operation because air density is lower in altitude. Based on a rule of thumb, for each 305 m above sea level, 3% performance losses occur at full load rating. Fig. 4.30 shows the typical effect of altitude on MTG output power.

When less than full power is required from a MT, the output is reduced by reducing rotational speed, which reduces temperature rise and pressure ratio through the compressor and temperature drop through the turbine,

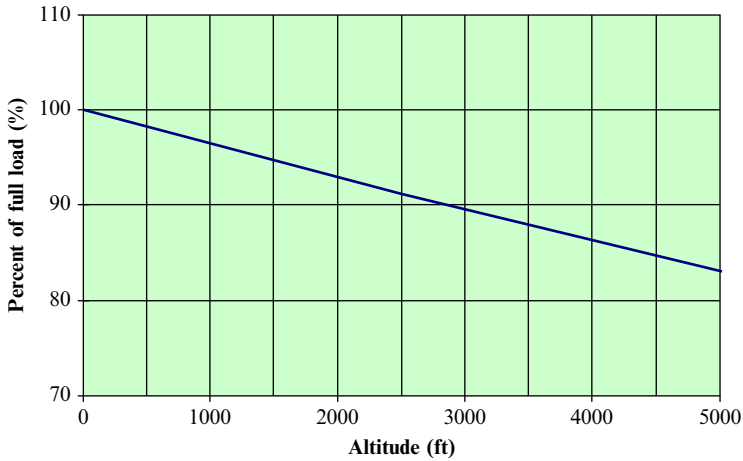


Fig. 4.30 Altitude effects on microturbine output power [10].

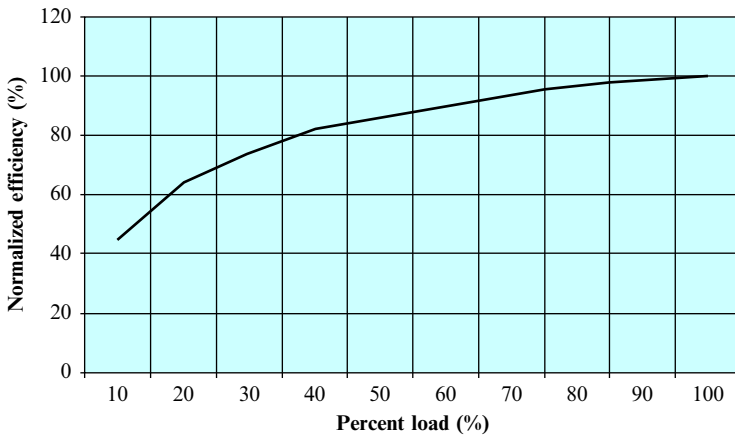


Fig. 4.31 Microturbine part-load efficiency [10].

and by reducing turbine inlet temperature so that the recuperator inlet temperature does not rise. In addition to reducing power, this change in operating conditions also reduces efficiency. The efficiency decrease is minimized by the reduction in mass flow (through speed reduction) at the same time as the turbine inlet temperature is reduced. Fig. 4.31 shows a typical part-load efficiency curve based on a 30 kW MT [10]. Experimental results show that when the MT is operated 50% below its rated capacity, pollutant emissions tend to increase.

MT operation in a partial load leads to lower efficiency. Efficiency as a function of output electrical power for two typical types of the MT is represented in Fig. 4.32.

Efficiency and output power as a function of ambient temperature for Capstone as the largest manufacture of MTs is shown in Figs. 4.33 through 4.35. Figs. 4.33–4.35 show C30, C65, and C200 Capstone MTs, respectively.

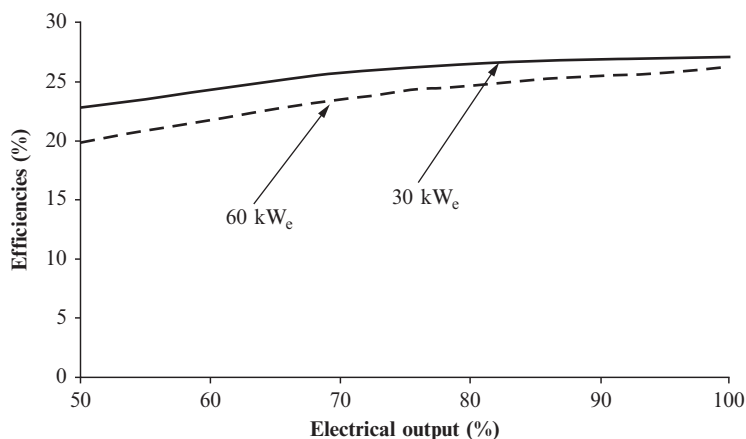


Fig. 4.32 Electrical efficiency for a 30 kW and a 60 kW microturbine [14].

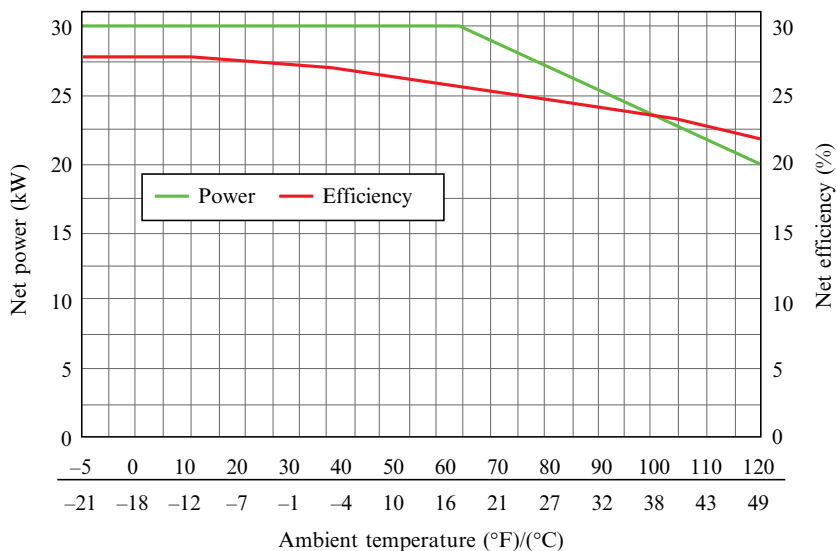


Fig. 4.33 The electrical power and net efficiency of a C30 MT as a function of temperature at sea level [9].

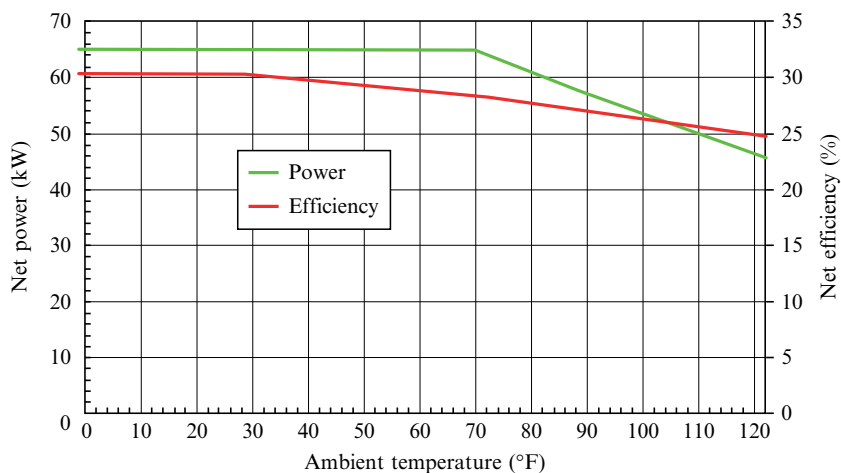


Fig. 4.34 The electrical power and net efficiency of a C65 MT as a function of temperature at sea level [9].

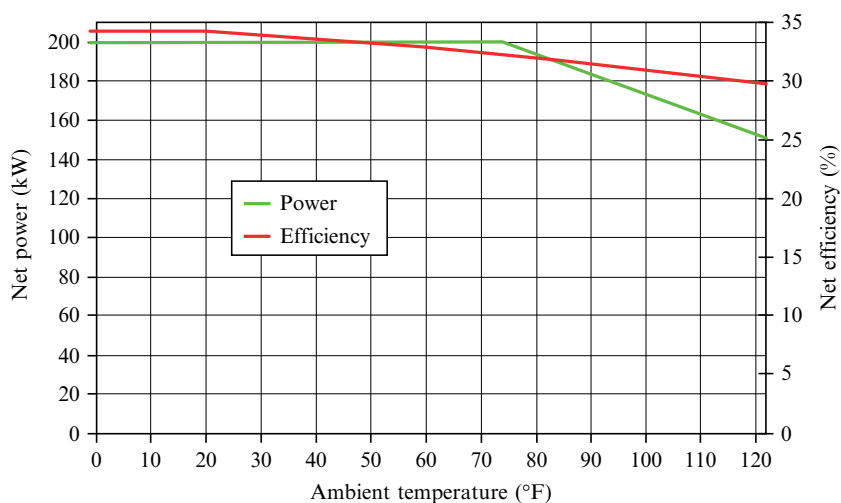


Fig. 4.35 The electrical power and net efficiency of a C200 MT as a function of temperature at sea level [9].

Evaluation of pollutant emission level is important issue and the MT with lower pollutant is more favorable. Manufacture and researchers are focused on more clean energy sources to prevent greenhouse gases emission. The US Environmental Protection Agency (EPA) lists six criteria air pollutants for

which ambient air limits have been set [15]. These air pollutants include NO_x , CO, SO_2 , lead (Pb), ozone (O_3) and particulates. And between them, NO_x , CO, and SO_2 are the most relevant for the operation of the MT. In the following, pollutant emission of the MTG is considered at various conditions.

Fig. 4.36 shows the pollutant emission characteristic of a 30 kW MT at steady state. MT operations at partial load release more pollutant compared to full load. Moreover, Fig. 4.37 shows the pollutant characteristic of a 60 kW MT at steady state.

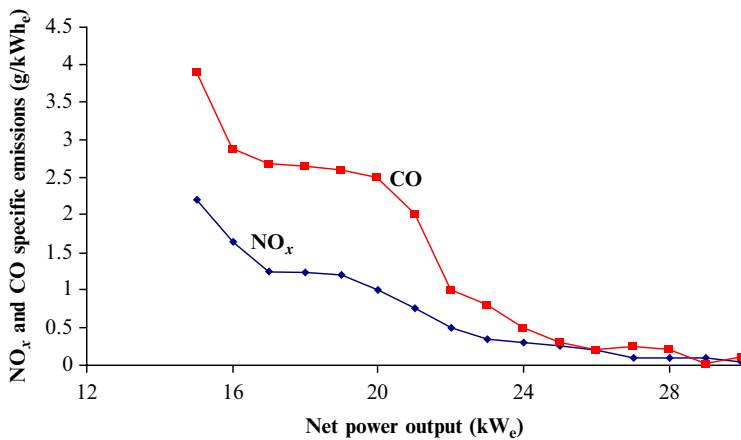


Fig. 4.36 NO_x and CO emission characteristic of the 30 kW MT [14].

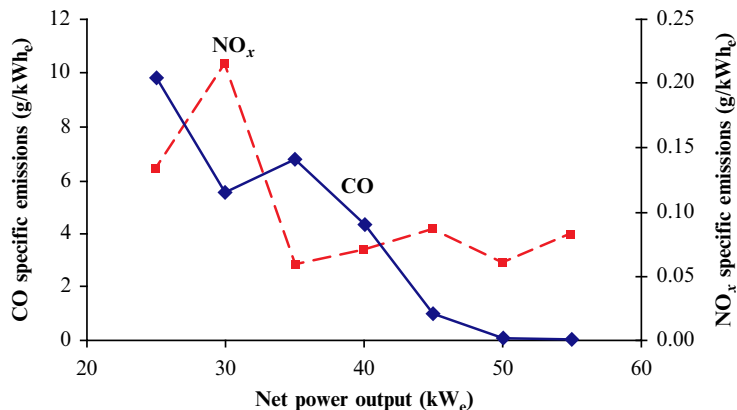


Fig. 4.37 NO_x and CO emission characteristic of the 60 kW MT [16].

Most studies about MT pollutant emissions are at steady state and full load conditions. The transient pollutant emissions during MT startup are often neglected. In fact, at partial loads and transient processes, the pollutant emission characteristics can be very different from rated conditions [12]. The following are from [12], and they reveal some results of the MT transient emissions startup at both partial and full load for various output powers. This study considers nitrogen oxide (NO_x), carbon monoxide (CO), and sulfur dioxide (SO_2) the most hazardous emissions for human health. In [12], a Turbec T100P MT fueled by natural gas used in the experimental studies of transient emissions. According to the manufacturer's catalog, output power of the unit at ISO conditions is 100 kW, and the efficiency is about 30%. The rated concentrations of CO and NO_x emissions at 15% oxygen are under 15 ppm.

Experimental results show that CO concentration of the MT startup increases rapidly to a rather high level (up to 5–100 times higher than that of the steady condition) and then drops down to a stable value [12]. There is no significant difference for various power outputs.

Moreover, the NO_x startup transient pollutant is similar to CO emission, and there is no significant difference in various powers. Generally speaking, the NO_x concentration at steady state condition is lower than 15 ppm, as stated in the manufacturer's catalog.

At higher power ratings, SO_2 emission is lower than 4 ppm, and it may be deduced that no significant damage occurs.

Finally, results of MT start-up pollutant emission show that [12]:

- Normal operation of the microturbine at full load power output produces the lowest emissions of air pollutants (CO, NO_x , and SO_2).
- The CO startup emissions have significant effect on the total amount of emission.
- Although the impact of NO_x and SO_2 startup emission is not as serious as that of CO, it also should be considered.

For comprehensive pollutant analysis, MT emissions compare with other DG technologies. The values are given for a variety of emissions, which are nitrogen oxide (NO_x), sulfur dioxide (SO_2), carbon dioxide (CO_2), and particulate matter (PM-10), and they are characterized in terms of pounds of emissions per unit of electrical output. The values are based on typical operation condition. Fig. 4.38 shows pollutant emission of the MT, which has favorable conditions in air emission. MT pollutant emission is desirable and catalytic exhaust gas cleaning is not required but natural gas fired reciprocating engines or diesel reciprocating engines are required.

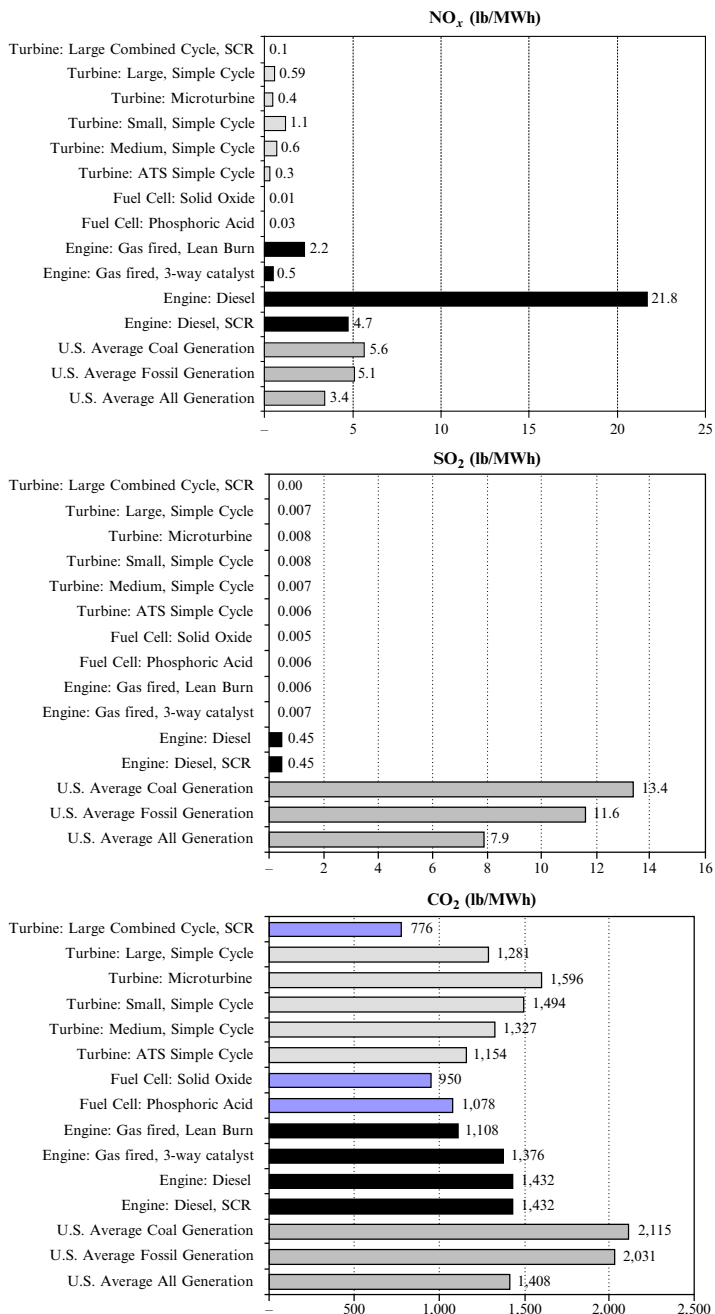


Fig. 4.38 Air emissions values for a number of distributed generation technologies [17].

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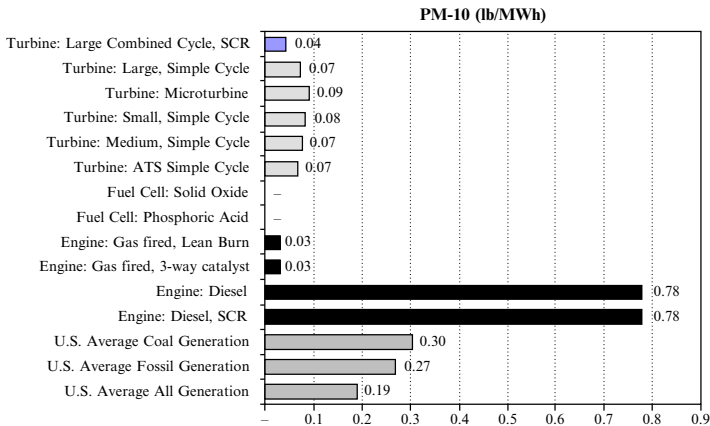


Fig. 4.38, cont'd

Although noise of the MT can be a problem, new technologies have solved this issue. Internal noises of the MT are from flow vortices, blading, magnetic field, and combustion. However, the level of noise is low and measurements show an average of 65 dB or less on a 10-m radius.

A single MT produces power in the range of 25–1000 kW, but research on higher output power is continuing and output power of MTs with parallel connections can reach up to 30 MW. Output power and efficiency of MTs are acceptable and appropriate in many cases. The efficiency of a non-recuperated MT is about 15%–20%, that for a recuperated MT is about 20%–35%, and in a cogeneration application or use as CHP efficiency can reach up to 80% [1].

Table 4.3 contains appropriate comparisons between various DGs. Based on Table 4.3 the MT as a reliable DG has a suitable rankings in many respects.

Table 4.3 Rankings for distributed power generating technologies [18]

Category	Most positive → least positive
Efficiency	Fuel cells—reciprocating—stirling—microturbine—rankine
Technology status	Reciprocating—microturbine—fuel cells—stirling—rankine
Cost	Reciprocating—microturbine—fuel cells—stirling—rankine
Emissions	Fuel cells—stirling—microturbine—rankine—reciprocating
Noise	Fuel cells—stirling—microturbine—rankine—reciprocating
Load matching flexibility	Reciprocating—fuel cells—microturbine—stirling—rankine

4.10 MT MODELING

The MT model includes temperature control, a fuel system, a speed governor, acceleration control, and turbine dynamic blocks. Outputs of the speed control, temperature control, and acceleration control are inputs of a low value selector (LVS), where the LVS output is the lowest value of all inputs and the fuel system is supplied by output of the LVS. A dynamic model of a single-shaft MT is presented in Fig. 4.39, and step-by-step modeling is done in the following sections. Note that parameters values are given in Appendix A to this chapter.

4.10.1 Speed and Acceleration Control

A speed controller operates on an error between reference speed and PMSG rotor speed, which is its output as it enters the LVS and usually a lead-lag transfer function or a proportional–integral–derivative (PID) controller is used for speed governor modeling. Fig. 4.40 shows a block diagram of the speed and acceleration control.

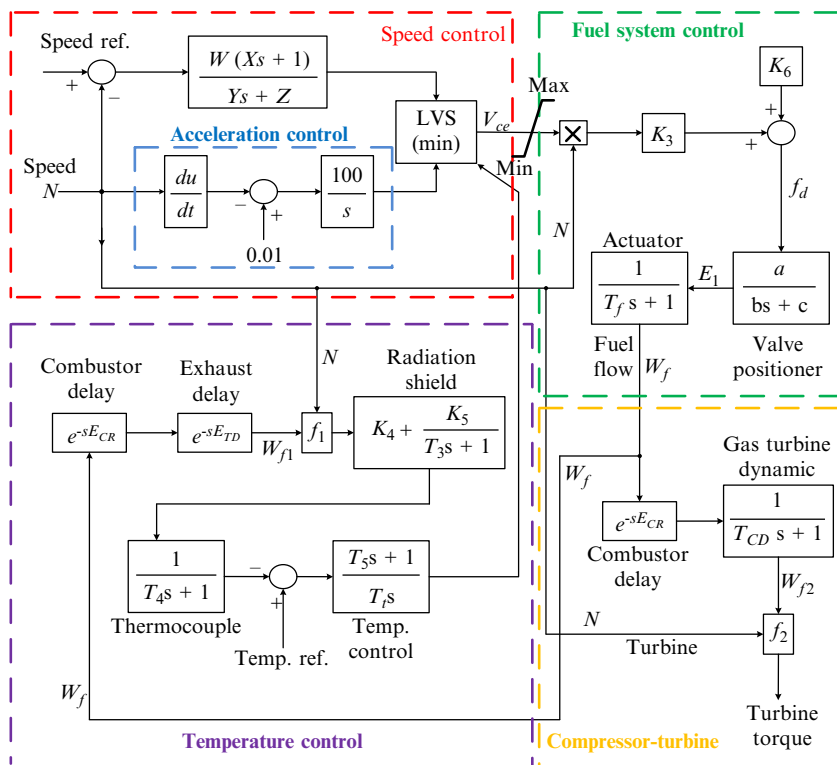


Fig. 4.39 Dynamic model of the microturbine.

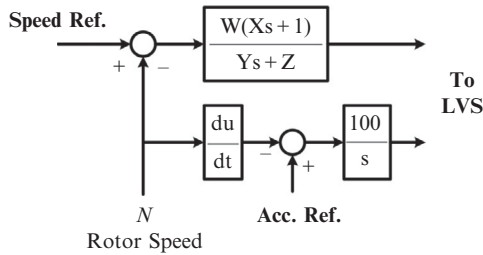


Fig. 4.40 Speed controller of a microturbine.

In this figure, W is controller gain, X (Y) is the governor lead (lag) time constant, and Z is a constant representing the governor mode in droop or isochronous [19].

Acceleration control is used for start-up time of the MT to limit the rate of rotor speed increasing before reaching operating speed. The system operating speed close to rated speed cause the elimination of the acceleration control in the modeling, but acceleration control is used in this study.

Example 4.7

Speed controller (governor) of a MT is modeled by a lead-lag transfer function with three parameters, which are X , Y , and Z . X and Y are the governor lead and lag time constant, respectively, and Z is a constant. Please illustrate the role of Z in the transfer function and the range of numbers that can be chosen for Z ?

Solution

Z is a constant that represents the governor mode in droop or isochronous. A droop governor is a straight proportional speed controller in which the output is proportional to the speed error. An isochronous speed controller is a proportional-plus-reset speed controller in which the rate of change of the output is proportional to the speed error. Therefore, the output of an isochronous governor will integrate in a corrective direction until the speed error is zero [20]. $Z=1$ represents a droop governor and $Z=0$ represents an isochronous governor.

4.10.2 Temperature Control

The input signals to the temperature control system are fuel demand signal and turbine speed, which output is a temperature control signal to the LVS. A temperature control block diagram is shown in Fig. 4.41.

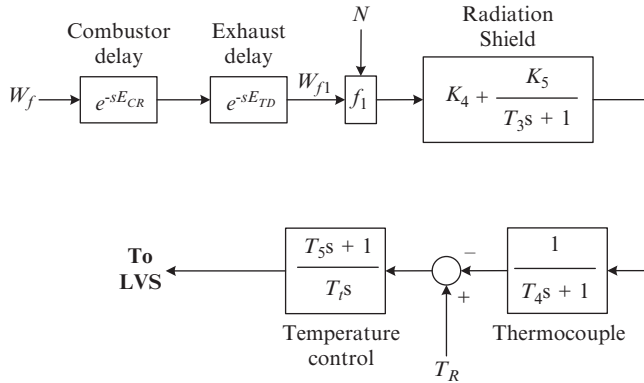


Fig. 4.41 Temperature controller of a microturbine.

The fuel and compressed air are burned in the combustor results in turbine torque. The exhaust temperature characteristic is as the following function:

$$f_1 = T_R - a_{f1} \cdot (1 - W_{f1}) + b_{f1} \cdot N \quad (4.14)$$

where f_1 is a function with inputs of fuel flow and turbine speed to produce a value of turbine exhaust temperature, a_{f1} and b_{f1} are constant values, and T_R is temperature reference. The exhaust temperature is measured using a series of thermocouples incorporating radiation shields [13]. According to Fig. 4.41, K_4 and K_5 are constant in radiation shield transfer function, T_3 and T_4 are the time constant of the radiation shield and thermocouple transfer function, respectively, and T_5 with T_t are the time constant of the temperature control transfer function. The output of the thermocouple is compared with a reference value and desirable performance is in high temperature. The reference temperature is normally higher than the thermocouple output but when thermocouple temperature is higher than reference value the result is a negative error, which is as input of the LVS and temperature control start decreasing to reach the former temperature.

4.10.3 Fuel System

Fuel system controls are a series block of the fuel valve and actuator. Fig. 4.42 shows a fuel control system for the MT. The output of the LVS, V_{ce} , is scaled by the gain K_3 and is offset by K_6 that is representing fuel flow at no load condition. The valve positioner transfer function is as follows:

$$E_l = \frac{a}{bs + c} f_d \quad (4.15)$$

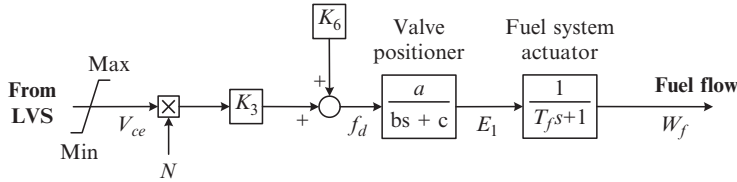


Fig. 4.42 Fuel control system of a microturbine.

Where a is the valve positioner gain, b is the valve positioner time constant, and f_d is input of the valve positioner. The actuator transfer function is

$$W_f = \frac{1}{T_f s + 1} E_l \quad (4.16)$$

where T_f is actuator time constant, E_l is output of the valve positioner, and W_f is the fuel demand signal per unit.

Example 4.8

Input of the MT fuel system is supply by output of the LVS block. The outputs of the speed controller, acceleration controller, and temperature controller enter the LVS and a minimum of them are passed through the LVS. Why is this mechanism chosen for the fuel system? What is the role of the limiter after the LVS?

Solution

These three control functions (speed governor, temperature control, and acceleration control) are all inputs to a LVS. The output of the LVS, which is called V_{ce} , is the lowest of the three inputs, whichever requires the least fuel. In other words, desirable operation of the MT depends on lower fuel consumption with higher output temperature and with respect to this fact, the LVS block and higher reference temperature are considered in modeling. Transfer from one control to another is bumpless and without any time lags. The output of the LVS is compared with maximum and minimum limits. Of the two, the maximum limit acts as a backup to temperature control and is not encountered in normal operation; the minimum limit is the more important, dynamically. This is because the minimum limit is chosen to maintain adequate fuel flow to ensure that a flame is maintained within the gas turbine combustion system [20].

Example 4.9

Output of the LVS in a MT fuel system is scaled by the gain K_3 and is offset by K_6 . What is the role of these parameters in the fuel system? Why are K_3 and K_6 equal to 0.77 and 0.23, respectively?

Solution

In terms of dynamic response, gas turbines have many differences from steam turbines. One of the more obvious differences is the need for a significant fraction of rated fuel to support self-sustaining, no-load conditions. This amounts to $\sim 23\%$ and is one of the economic driving forces to minimize operating time at full-speed, no-load conditions [20]. To allow the use of governor parameters comparable to steam turbines and hydroturbines, the governor operates from 0%–100% of the active load range. This range must then be corrected to 23%–100% of the fuel flow range to be compatible with the thermodynamics of the turbine. This relationship between governor output signal and fuel flow is shown in Fig. 4.43. With respect to the mentioned explanation, K_6 is equal to 0.23 and that represents fuel flow at a no-load condition and $K_3 = 1 - K_6 = 1 - 0.23 = 0.77$.

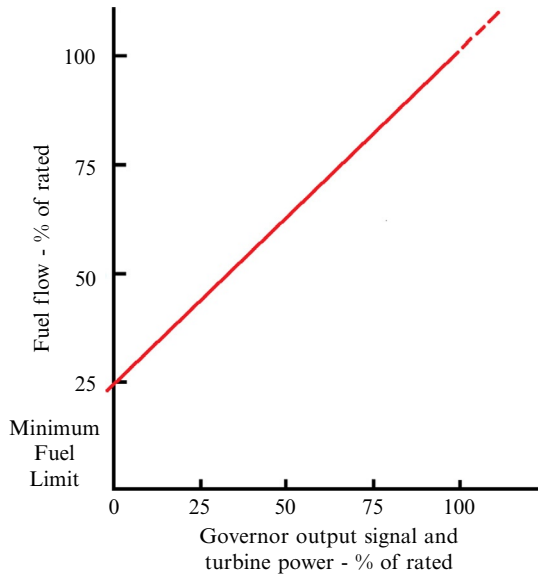


Fig. 4.43 Relationship between governor output signal and turbine fuel flow.

4.10.4 Compressor-Turbine

The compressor-turbine package is an important part of a MT and they are considered as a package because they are mounted on the same shaft. The input signals to the gas turbine are the fuel flow W_f signal that is achieved from the fuel control system and the rotor speed deviation ΔN . The compressor-turbine package is represented in Fig. 4.44. The output signal is the turbine torque. The gas turbine dynamic transfer function is

$$W_{f2} = \frac{1}{T_{CD}s + 1} W_f \quad (4.17)$$

where T_{CD} is the gas turbine dynamic time constant.

The torque characteristic is as following equation:

$$f_2 = a_{f2} + b_{f2} \cdot W_{f2} + c_{f2} \cdot N \quad (4.18)$$

where a_{f2} , b_{f2} , and c_{f2} are constant values and f_2 is a function whose inputs are fuel flow and turbine speed to produce mechanical power.

Note that all mentioned controllers are per unit except the temperature controller. Temperature control mostly is based on English units with $T_R = 950^\circ\text{F}$.

4.10.5 PMSG Model

Electric power is produced by using a high-speed generator, which is driven by a turbine. A PMSG is similar to the conventional alternator configuration, and modeling of the PMSG is expressed in rotor reference frame (d - q frame). The following equations are applied for transformation of three-phase abc to rotor reference frame $dq0$ and vice versa.

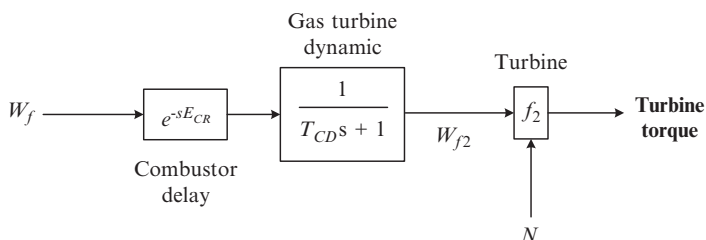


Fig. 4.44 Compressor-turbine package of a microturbine.

$$\begin{bmatrix} v_q \\ v_d \\ v_o \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta_r & \cos(\theta_r - 120) & \cos(\theta_r + 120) \\ -\sin \theta_r & -\sin(\theta_r - 120) & -\sin(\theta_r + 120) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \quad (4.19)$$

$$\begin{bmatrix} v_d \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} \cos \theta_r & -\sin \theta_r & 1 \\ \cos(\theta_r - 120) & -\sin(\theta_r - 120) & 1 \\ \cos(\theta_r + 120) & -\sin(\theta_r + 120) & 1 \end{bmatrix} \begin{bmatrix} v_q \\ v_d \\ v_o \end{bmatrix} \quad (4.20)$$

And then PMSG modeling in d - q form are given by Eqs. (4.21)–(4.23).

- *Electrical equations*

$$v_d = R_s i_d + L_d \frac{di_d}{dt} - L_q p \omega_r i_q \quad (4.21)$$

$$v_q = R_s i_q + L_q \frac{di_q}{dt} + L_d p \omega_r i_d + \lambda p \omega_r \quad (4.22)$$

$$T_e = 1.5P [\lambda i_d + (L_q - L_d) i_d i_q] \quad (4.23)$$

where, L_d and L_q are the d -axis and q -axis inductance, respectively; R_s is the stator winding resistance; i_d and i_q are the d -axis and q -axis current, respectively; v_d and v_q are the d -axis, q -axis voltage, respectively; ω_r is the angular velocity of the rotor; λ is the flux linkage; P is the number of poles; and T_e is the electromagnetic torque.

- *Mechanical equations*

$$\frac{d\omega_r}{dt} = \frac{1}{J} (T_e - T_{shaft} - F\omega_r) \quad (4.24)$$

$$\frac{d\theta_r}{dt} = \omega_r \quad (4.25)$$

where J is rotor and load combined inertia, T_{shaft} is mechanical torque, F is rotor and load combined viscous friction, and θ_r is the rotor angular position.

Fig. 4.45 shows the rotor reference frame equivalent circuit of a PMSG model.

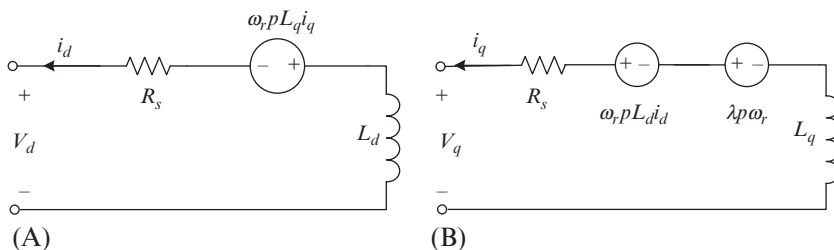


Fig. 4.45 Permanent magnet synchronous generator equivalent circuit model: (A) d -axis and (B) q -axis.

Example 4.10

A 30 kW, three-phase, PMSG produces an open-circuit voltage of 400 V line-to-line, 60 Hz, when driven at a speed of 96,000 rpm. When operating at rated speed and supplying a resistive load, its terminal voltage is observed to be 320 V line-to-line for a power output of 15 kW.

- Calculate the generator phase current under this operating condition.
- Assuming the generator armature resistance to be negligible, calculate the generator 60 Hz synchronous reactance.
- Write the torque in normal three-phase (abc) frame and compare with Eq. (4.23).
- What is the difference between permanent magnet synchronous motor and generator?

Solution

- Because the load is resistive,

$$I_a = \frac{P}{3V_a} = \frac{15,000}{\sqrt{3} \times 320} = 27.06 \text{ A}$$

-

$$E_{af} = 400 / \sqrt{3} = 230.9 \text{ V}$$

So we have,

$$E_{af} = \sqrt{V_a^2 - (X_s I_a)^2} \rightarrow X_s = \frac{\sqrt{E_{af}^2 + V_a^2}}{I_a} = 14.58 \Omega$$

- We assume that linkage flux is sinusoidal:

$$\lambda_m = \lambda_{m0} \begin{bmatrix} \sin \theta_r \\ \sin \left(\theta_r - \frac{2\pi}{3} \right) \\ \sin \left(\theta_r + \frac{2\pi}{3} \right) \end{bmatrix}$$

where λ_{m0} is amplitude of λ_m . According to the mentioned equation for linkage flux, the stator-induced voltages are sinusoidal. The torque standard equation in electrical machines is

$$T_e = \frac{p}{2} \frac{\partial W_c}{\partial \theta_r}$$

where W_c is coenergy and p is the number of poles. The coenergy equation is as follows: $W_c = \frac{1}{2} i_{abc}^T L_s i_{abc} + i_{abc}^T \lambda_m + W_{pm}$

where L_s is stator inductance and W_{pm} is the energy of the coupling field. From the previously mentioned torque equation and coenergy equations we have

$$T_e = \pm \left(\frac{p}{2} \right) \left\{ \frac{(L_{md} - L_{mq})}{3} \left[\left(i_{as}^2 - 0.5i_{bs}^2 - i_{as}i_{cs} + 2i_{bs}i_{cs} \right) \sin 2\theta_r + \frac{\sqrt{3}}{2} \left(i_{bs}^2 - i_{cs}^2 - 2i_{as}i_{bs} + 2i_{as}i_{cs} \right) \cos 2\theta_r \right] + \lambda_{m0} \left[(i_{as} - 5i_{bs} - 0.5i_{cs}) \cos \theta_r + \frac{\sqrt{3}}{2} (i_{bs} - i_{cs}) \sin \theta_r \right] \right\}$$

where $L_{md} = 3/2(L_A - L_B)$ and $L_{mq} = 3/2(L_A + L_B)$

- (d) To illustrate the PMSG it is easier to focus on motor type. The equations are similar but there are two great differences. Current direction in the generator type is outgoing and the electromagnetic torque sign is negative. So, equations and equivalent circuit are based on permanent magnet generator and permanent magnet motor is similar with the mentioned difference. You can see [21] for better understanding.

4.11 CONTROL CIRCUITS

A power conditioning unit consists of a rectifier-inverter system with a DC link. It is a general configuration of a power electronic interface in MT units (MTUs). The purposes of this configuration are simple structure, simple control circuit, and ease of understanding. The high-frequency electric power of a PMSG must be converted to DC and inverted back to 60 or 50 Hz AC. An insulated gate bipolar transistor (IGBT)-based pulse width modulation (PWM) inverter is used with a 2 kHz carrier frequency. The inverter injects AC power from a DC link of the MTU to the AC distribution system [22,23]. Grid-connected mode allows the MTU to operate parallel to the grid, providing base loading and peak shaving and grid support. Moreover, stand-alone mode allows the MTU to operate completely isolated from the grid. There are two different control strategies for each operation mode [24,25]:

- P-Q control strategy for grid-connected (power control)
- V-f control strategy for stand-alone (voltage-frequency control)

4.11.1 Control Strategy for Grid-Connected Operation

In this situation, voltage amplitude and frequency of the MTG is determined by the grid. The grid is like an infinite bus with constant frequency and voltage in which appropriate power flow is an important issue. In other words, for grid-connected operation mode, a P-Q control strategy is used where delivering desired active and reactive instantaneous power to load is

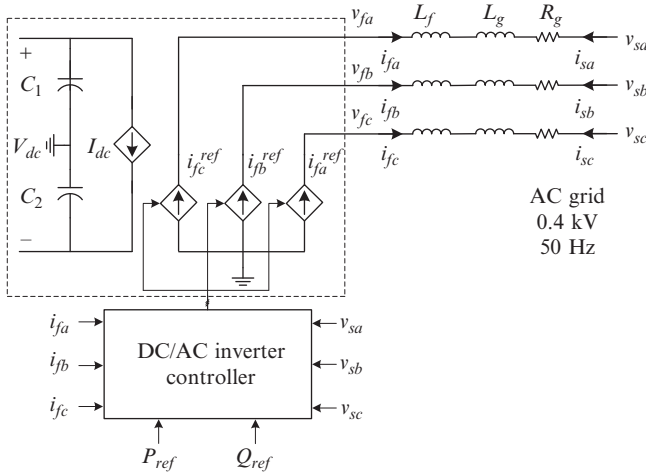


Fig. 4.46 Equivalent circuit of the inverter in grid-connected mode.

considered and the power deficit is compensated by the grid. The equivalent circuit of a DC/AC inverter in grid-connected operating condition is shown in Fig. 4.46. This converter is represented with three ideal current sources, which are i_{fa}^{ref} , i_{fb}^{ref} , and i_{fc}^{ref} . The converter manages the amount of the current injected to the AC grid from the DC bus, and the inverter controller uses the hysteresis current control (HCC) switching technique. The input signals to the inverter control circuit are three-phase grid voltages, which are called v_{sa} , v_{sb} , and v_{sc} , three-phase output currents of the converter, which are called i_{fa} , i_{fb} , and i_{fc} , reference of the active power P_{ref} and the reference of the reactive power Q_{ref} . With respect to Fig. 4.46, L_f is inductance of the converter filter, and R_g and L_g are the resistance and inductance of the grid, respectively.

Fig. 4.47 shows the inverter control strategy in grid-connected mode. The required power that must be injected to the grid is determined by P_{ref} and Q_{ref} signals, which are set by power management units. Based on Fig. 4.47, equations are expressed in Clarke transformation or α - β frame:

$$\begin{bmatrix} i_{sa} \\ i_{sb} \\ i_{sc} \end{bmatrix} = - \begin{bmatrix} i_{fa} \\ i_{fb} \\ i_{fc} \end{bmatrix} \quad (4.26)$$

$$\begin{Bmatrix} v_{s\alpha} \\ v_{s\beta} \\ v_{s0} \end{Bmatrix} = T_{\alpha\beta 0} \begin{Bmatrix} v_{sa} \\ v_{sb} \\ v_{sc} \end{Bmatrix}, \quad T_{\alpha\beta} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix} \quad (4.27)$$

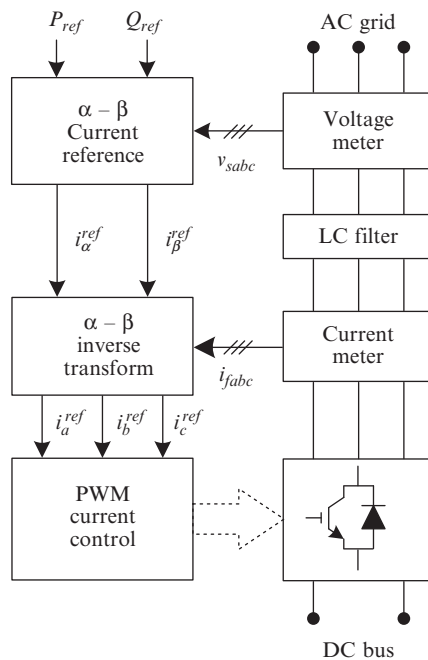


Fig. 4.47 Control strategy of the inverter in grid-connected mode.

With Eq. (4.27), the voltages and currents convert to α - β frame. Note that in a balanced three-phase system, the third equation is equal to zero. The instantaneous active and reactive power are as in Eqs. (4.28) and (4.29), respectively:

$$P = \frac{3}{2} (v_{\alpha} i_{\alpha} + v_{\beta} i_{\beta}) \quad (4.28)$$

$$Q = \frac{3}{2} (v_{\beta} i_{\alpha} - v_{\alpha} i_{\beta}) \quad (4.29)$$

The α - β reference currents are related to the reference power by Eq. (4.30).

$$\begin{bmatrix} i_{s\alpha}^{ref} \\ i_{s\beta}^{ref} \end{bmatrix} = \frac{2/3}{v_{s\alpha}^2 + v_{s\beta}^2} \begin{bmatrix} v_{s\alpha} & v_{s\beta} \\ v_{s\beta} & -v_{s\alpha} \end{bmatrix} \begin{bmatrix} P_{ref} \\ Q_{ref} \end{bmatrix} \quad (4.30)$$

HCC requires three-phase current references, which are achieved by α - β inverse transformation.

$$\begin{bmatrix} i_{sa}^{ref} \\ i_{sb}^{ref} \\ i_{sc}^{ref} \end{bmatrix} = T_{\alpha\beta 0}^{-1} \begin{bmatrix} i_{s\alpha}^{ref} \\ i_{s\beta}^{ref} \\ i_{s0}^{ref} \end{bmatrix}, \quad T_{\alpha\beta 0}^{-1} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \quad (4.31)$$

$$\begin{bmatrix} i_{fa}^{ref} \\ i_{fb}^{ref} \\ i_{fc}^{ref} \end{bmatrix} = - \begin{bmatrix} i_{sa}^{ref} \\ i_{sb}^{ref} \\ i_{sc}^{ref} \end{bmatrix} \quad (4.32)$$

Finally, inverter switching is done through comparison between calculated reference currents and inverter actual measurement currents.

4.11.2 Control Strategy for Stand-Alone Operation

In this situation, voltage amplitude and frequency are important because the MTG cannot automatically set these parameters. The equivalent circuit of the inverter in islanding mode is represented in Fig. 4.48.

The inverter is represented with three ideal voltage sources, which are called v_{fa}^{ref} , v_{fb}^{ref} , and v_{fc}^{ref} , to regulate load voltages. Input signals to the inverter control circuit are load voltages and desired frequency.

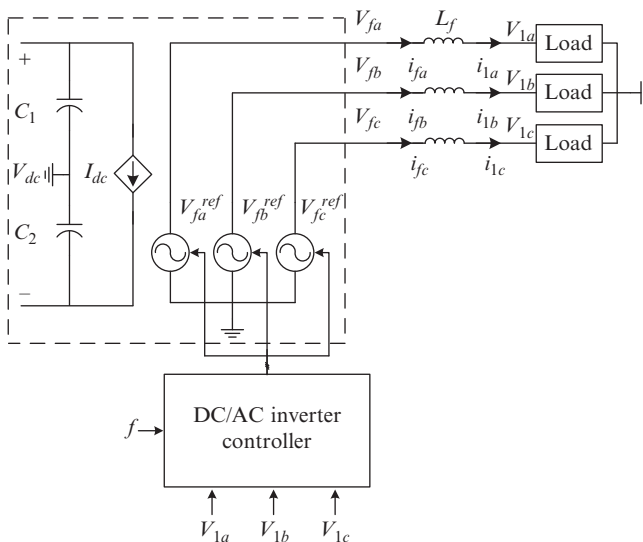


Fig. 4.48 Equivalent circuit of the inverter in stand-alone mode.

The following equation describes inverter voltages and currents:

$$\begin{bmatrix} v_{fa} \\ v_{fb} \\ v_{fc} \end{bmatrix} = \begin{bmatrix} v_{la} \\ v_{lb} \\ v_{lc} \end{bmatrix} + \begin{bmatrix} L_f & 0 & 0 \\ 0 & L_f & 0 \\ 0 & 0 & L_f \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_{fa} \\ i_{fb} \\ i_{fc} \end{bmatrix} \quad (4.33)$$

where v_{fa} , v_{fb} , and v_{fc} are three-phase output voltage of the inverter; i_{fa} , i_{fb} , and i_{fc} are three-phase output currents; v_{la} , v_{lb} , and v_{lc} are three-phase load voltages; and L_f is inductance of the output filter. Based on Eq. (4.19), the voltage formula in Eq. (4.33) converts to Park transformation or $dq0$ reference frame as follows:

$$\begin{bmatrix} v_{fd} \\ v_{fq} \\ v_{f0} \end{bmatrix} = \begin{bmatrix} v_{ld} \\ v_{lq} \\ v_{l0} \end{bmatrix} + \begin{bmatrix} L_f & 0 & 0 \\ 0 & L_f & 0 \\ 0 & 0 & L_f \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_{fd} \\ i_{fq} \\ i_{f0} \end{bmatrix} + \begin{bmatrix} 0 & -\omega L_f & 0 \\ \omega L_f & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{fd} \\ i_{fq} \\ i_{f0} \end{bmatrix} \quad (4.34)$$

The circuit configuration and control scheme for the inverter in islanding mode is depicted in Fig. 4.49. In this mode, V-f control strategy is used so that voltage magnitude and frequency are adjustable parameters. In the V-f controller, desirable frequency (e.g., 50 Hz) can be obtained by a phase-locked loop (PLL) as well as desired load voltages set by a proportional integral (PI) controller to eliminate steady state errors.

The output filter is used to eliminate harmonics and ripples to achieve sinusoidal output. At first, load voltages measure and convert to $dq0$ frame as follows:

$$\begin{bmatrix} v_{ld} \\ v_{lq} \\ v_{l0} \end{bmatrix} = T_{dq0} \begin{bmatrix} v_{la} \\ v_{lb} \\ v_{lc} \end{bmatrix}, \quad T_{dq0} = \frac{2}{3} \begin{bmatrix} \cos(\omega t) & \cos(\omega t - 120) & \cos(\omega t + 120) \\ -\sin(\omega t) & -\sin(\omega t - 120) & -\sin(\omega t + 120) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (4.35)$$

Note that in balanced three-phase voltages, v_{l0} is equal to zero. After converting to $dq0$ frame, it is required to achieve the difference between expected voltage and converted measured voltage because the load phase voltage should be kept balanced and sinusoidal with constant amplitude and frequency. So, the expected load voltages per unit are considered as the following values:

$$\begin{bmatrix} v_{ldp} \\ v_{lqp} \\ v_{l0p} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (4.36)$$

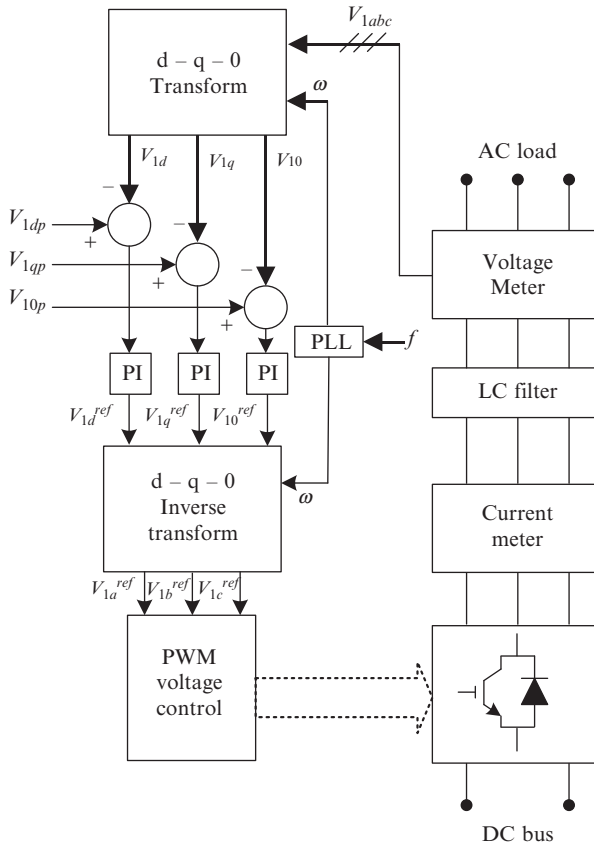


Fig. 4.49 Control strategy of the inverter in stand-alone mode.

Reference voltages of the inverter conclude from the difference between expected values and measured values after passing through a PI controller. The reference voltages in $dq0$ rotating frame is as in Eq. (4.37):

$$\begin{bmatrix} v_{1d}^{ref} \\ v_{1q}^{ref} \\ v_{10}^{ref} \end{bmatrix} = \begin{bmatrix} \text{PI}(v_{1dp} - v_{1d}) \\ \text{PI}(v_{1qp} - v_{1q}) \\ \text{PI}(v_{10p} - v_{10}) \end{bmatrix} \quad (4.37)$$

Inverse transform of the inverter voltages cause the desired voltage for switching to be achieved.

$$\begin{bmatrix} v_{1a}^{ref} \\ v_{1b}^{ref} \\ v_{1c}^{ref} \end{bmatrix} = T_{dq0}^{-1} \begin{bmatrix} v_{1d}^{ref} \\ v_{1q}^{ref} \\ v_{10}^{ref} \end{bmatrix}, \quad T_{dq0}^{-1} = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) & 1 \\ \cos(\omega t - 120) & -\sin(\omega t - 120) & 1 \\ \cos(\omega t + 120) & -\sin(\omega t + 120) & 1 \end{bmatrix} \quad (4.38)$$

The reference output voltages for the inverter are transformed to the abc by using Eq. (4.38), and then the available voltages in the abc coordinate are compared with the triangular wave provided by the PWM voltage control block and then apply suitable switching pattern to the inverter.

4.12 SIMULATION RESULTS

To analyze the previously mentioned structure as well as the impacts of a MTG unit on the distribution system, simulation is carried out in PSCAD/EMTDC software. The MTG system that is considered in the simulation is shown in Fig. 4.50. The power electronic interface is based on AC/DC/AC through a passive rectifier and the system parameters are given in Appendix A to this chapter. The simulation scenarios are focused on suitable operation with supplying unbalanced passive loads in the grid-connected and islanding modes. All time functions are in seconds and speed reference is kept constant at 1 p.u.

4.12.1 Grid-Connected Operation

In this case, the P-Q control scheme is applied. The reference values of P_{ref} and Q_{ref} are changed from 8 to 16 kW and from 1.4 to 3.5 kVar, at $t = 10$ s, and then back to 8 kW and 1.4 kVar, at $t = 20$ s, respectively.

Fig. 4.51 shows P and Q delivered to the grid and the system output power is varied according to the reference points.

Fig. 4.52 shows the rotor speed variations (N) and fuel demand signal (W_f). Rotor speed set at 0.988 p.u. after transient state variations, and a negative change occurs along with power variations. When power is increased, speed is decreased to 0.977 p.u. and then back again to initial value with power decreasing. In Fig. 4.52B, when the power reference variations are applied,

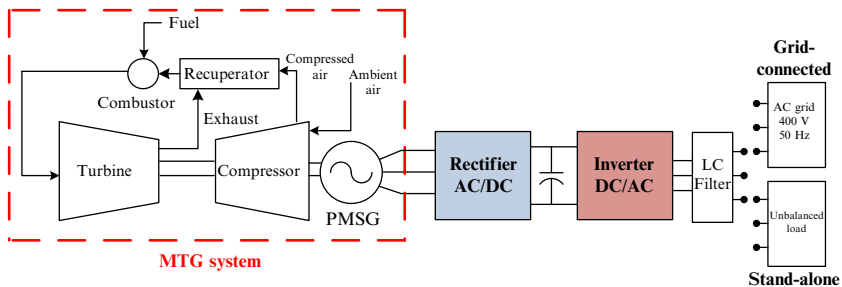


Fig. 4.50 Microturbine generation system.

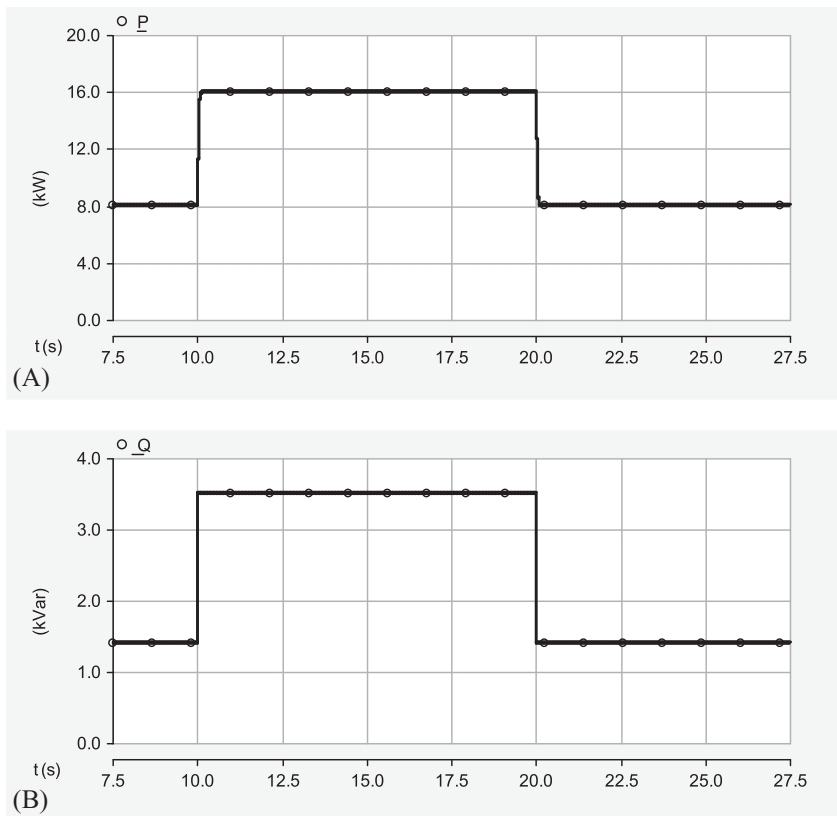


Fig. 4.51 Power injected to the grid: (A) active power and (B) reactive power.

fuel demand for combustion process is increased rapidly from 0.45 p.u. at 8 kW to 0.67 p.u. at 16 kW and vice versa. According to Fig. 4.52, fuel demand variations are the opposite of rotor speed variations.

Fig. 4.53A shows the shaft torque (T_{shaft}) and the developed electromagnetic torque (T_e). The shaft torque and electromagnetic torque are about equal at steady state. Fig. 4.53B shows the turbine mechanical torque with delay caused by time constants, which are considered in the model of the turbine and it follows the variations of the reference values.

Fig. 4.54A shows the power output from the MT and DC bus, which are equal at steady state. The DC bus voltage of the diode rectifier is shown in Fig. 4.54B. Note that the DC bus voltage drops when the load is increased and returns to its initial value, which is about 785 V when power is at its initial level.

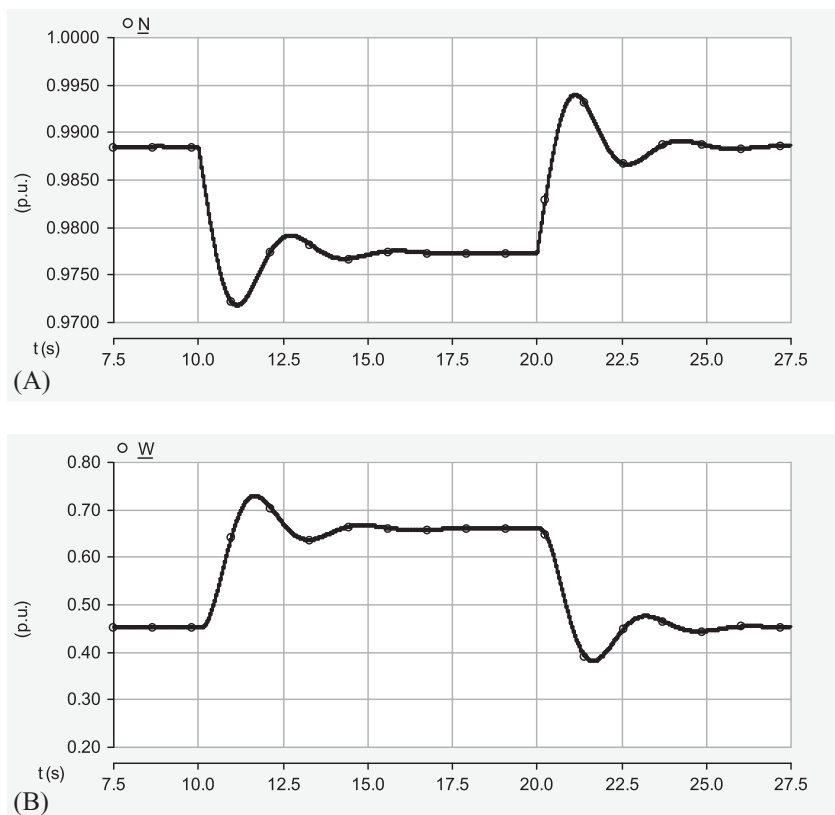


Fig. 4.52 (A) Rotor speed and (B) fuel demand signal of the microturbine in grid-connected mode.

Fig. 4.55 shows grid-side phase voltages and line currents of the inverter output. The line currents change with power reference variations. However, the inverter can deliver the MTGs power to the grid with low harmonic current. This verifies the effectiveness of the P-Q control strategy.

4.12.2 Stand-Alone Operation

In this mode, MTG response to unbalanced load in the stand-alone operation mode is studied. Load parameters are given in Appendix A to this chapter and V-f control strategy is used. The unbalanced load No. 1 is changed to the unbalanced load No. 2 at $t = 10$ s and the load is reverse changed to its initial value at $t = 20$ s.

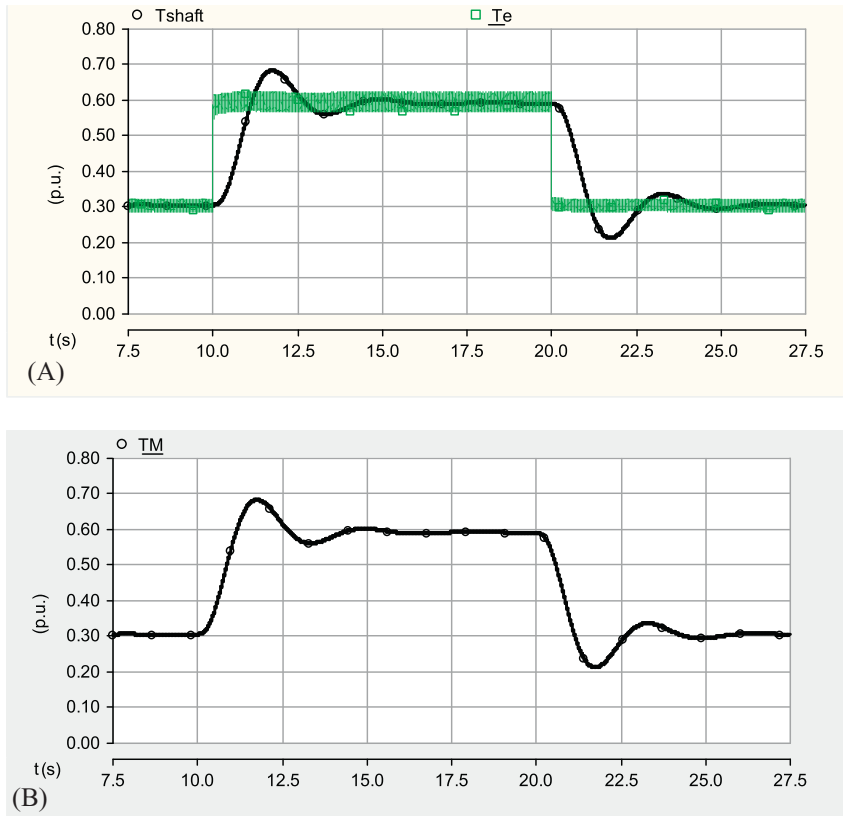


Fig. 4.53 (A) Shaft torque and electric torque and (B) turbine mechanical torque of the microturbine in grid-connected mode.

Fig. 4.56 shows the active and reactive power consumed by unbalanced AC loads. When the load is increased, power output is increased, too, and vice versa. Fig. 4.57 shows the MTG rotor speed variations and fuel signal in supplying unbalanced loads. Similar to what was mentioned earlier, when demand is increased this causes speed to drop and the fuel consumed to increase, and when demand is decreased it leads to an increase of rotor speed and a decrease of fuel demand signal. When the MT is supplied the unbalanced AC loads No. 1 and No. 2, the rotor speed is equal to 0.988 and 0.977 p.u. respectively. When at $t = 10$ s, load No. 2 is applied, the fuel demand for combustion process is increased rapidly to 0.661 p.u. from 0.458 p.u. and vice versa.

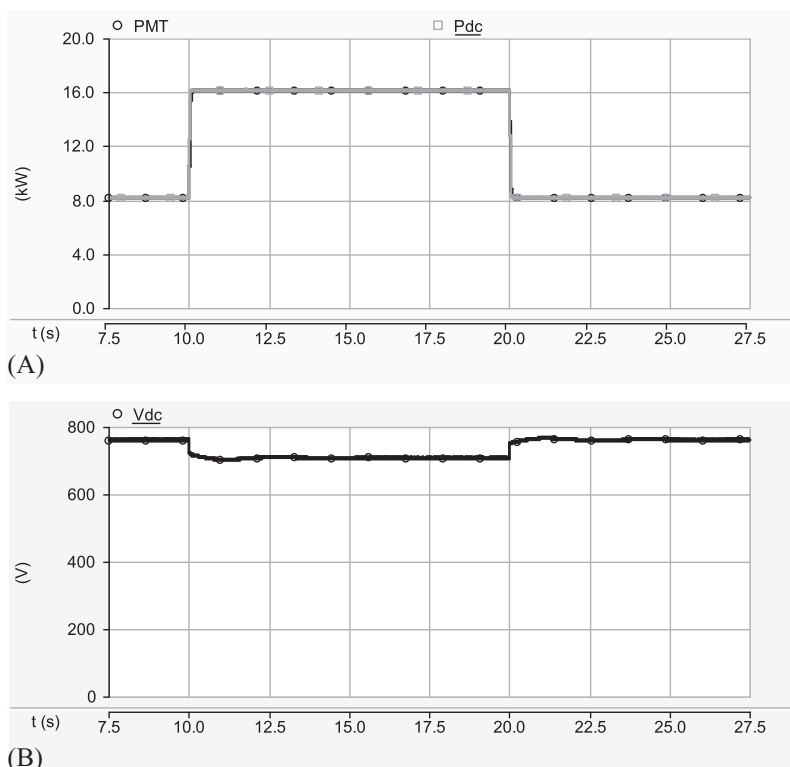


Fig. 4.54 (A) Microturbine and DC bus power output and (B) DC bus voltage in grid-connected mode.

The shaft torque (T_{shaft}) and electromagnetic torque (T_e) are shown in Fig. 4.58A. The shaft torque and electromagnetic torque are equal together at steady state. Fig. 4.58B shows the turbine mechanical output torque, which is matched according to load variations.

Fig. 4.59A shows the power output from the MT and DC bus, which is matched with unbalanced loading conditions. Fig. 4.59B shows the DC bus voltage of the uncontrolled rectifier, when the unbalanced loads are connected to the MT system. The DC bus voltage fluctuates and drops back to initial value, when No. 2 is disconnected from the MTG.

Fig. 4.60 shows the phase voltages and line currents at unbalanced load terminals. The inverter maintains the output at the desired level irrespective of the load applied on the system. The AC voltage level across the load

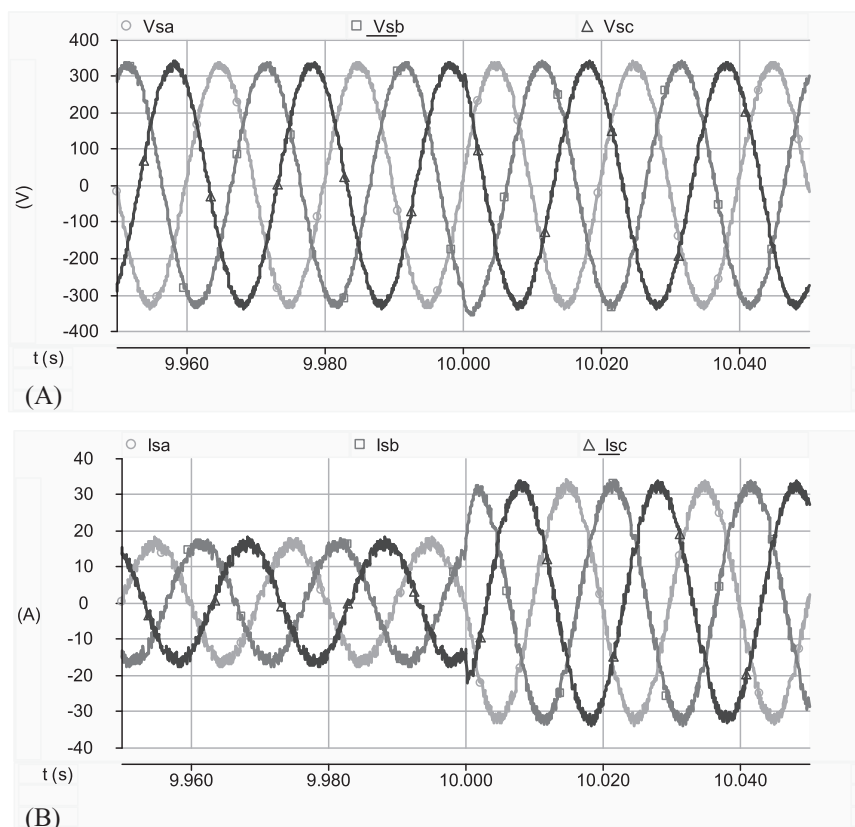


Fig. 4.55 (A) Grid-side phase voltages and (B) line currents of the inverter in grid-connected mode.

remains unchanged with the load variation switching and staying at the reference value the whole time of simulation. The balanced voltages are provided for unbalanced AC loads, while the load phase currents are not sinusoidal and balanced. To quantify level of the voltage unbalance, the percentage of unbalance is expressed by “degree of unbalance in three-phase system” [26]. In this case, the negative sequence unbalance is lower than 1.1%, which is acceptable. It should be noted that international standards admit unbalances lower than 2% [26]. The three-phase line currents at load terminals change with the load variation switching. It should be noted that the frequency of output voltage of the PMSG is over 1.6 kHz, while that at the output of the inverter, at the load terminals,

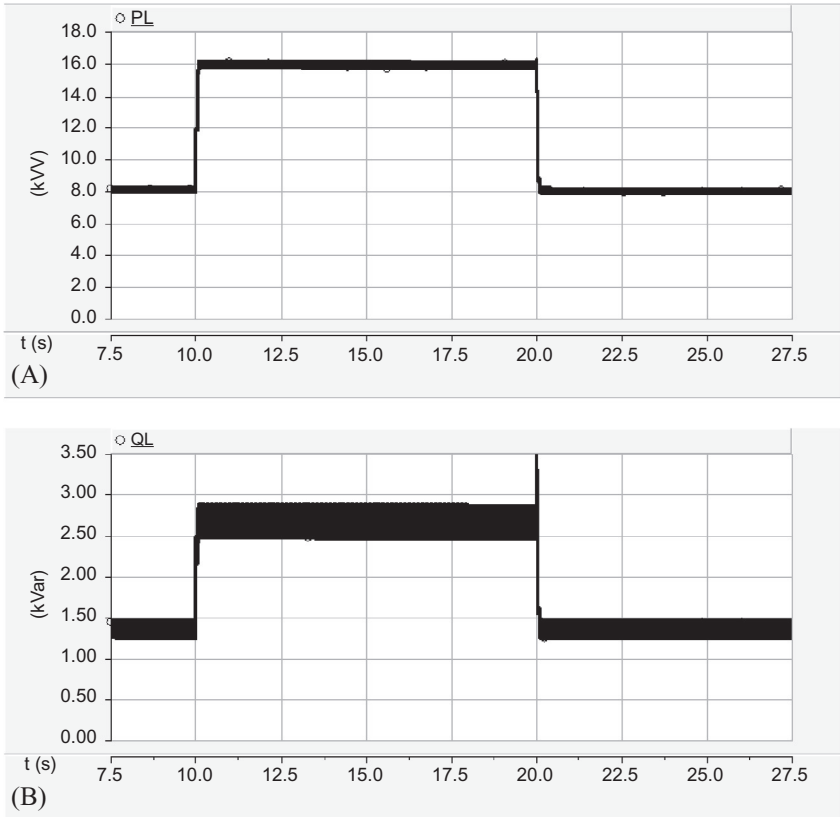


Fig. 4.56 Unbalanced loads power consumed: (A) active power and (B) reactive power.

it is 50 Hz. This verifies the effectiveness of the V-f control strategy for the stand-alone operation.

4.13 FUTURE TRENDS

Considering MTs potential advantages, a bright future can be expected for them. MTGs are suitable for DG systems because of their low maintenance and high reliability. One characteristic that drives manufacturers to focus on them is efficiency improvement. A MT with higher efficiency and fuel flexibility is an ideal choice for power generation. Higher efficiency of the MT in renewable fuels like biogas is another subject that will be considered. Further size reduction is another issue that is considered by manufacturers and

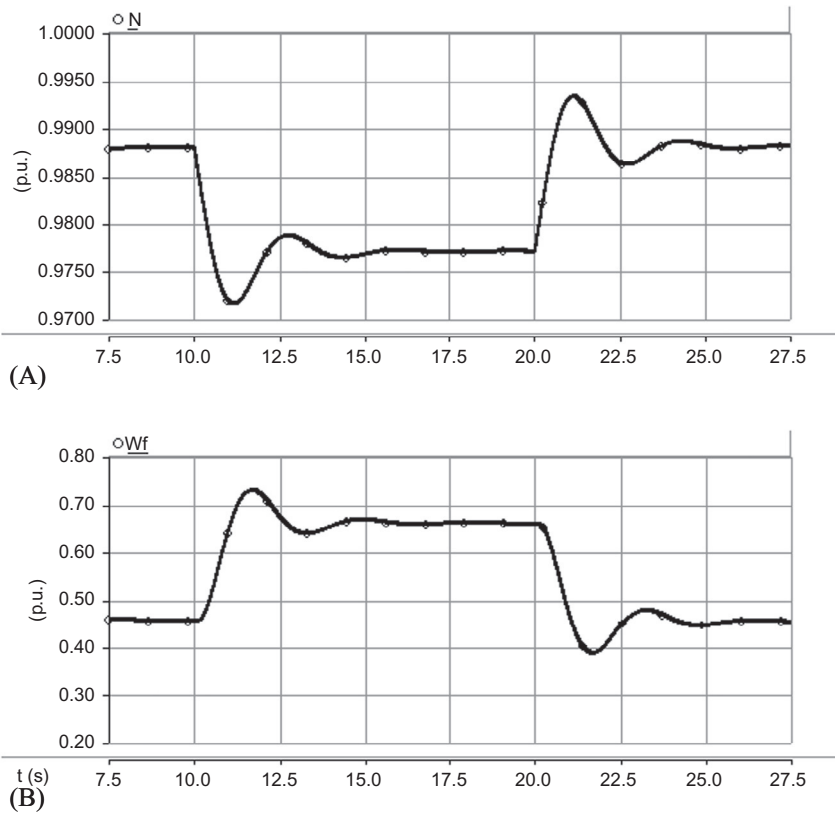


Fig. 4.57 (A) Rotor speed and (B) fuel demand signal of the microturbine in stand-alone mode.

extensive research is being done. Higher efficiency and smaller size cause more global attention to MTGs.

Recently, a combination of two or more DGs is considered for maximum use of sources and MT has an important role in this regard. The combination of a MT with a fuel cell is one of the best sources; however, other combinations such as CCHP with PV are also considered. With the advancement of technology, using a single DG is not desirable.

Global attention on reduction of emissions like CO_2 and NO_x and achieving higher efficiency are major reasons for expanding the use of hybrid vehicles. Using MTs in hybrid vehicles and transportation as well as aircraft, turbojets, turboprops, and unmanned vehicles are developing.

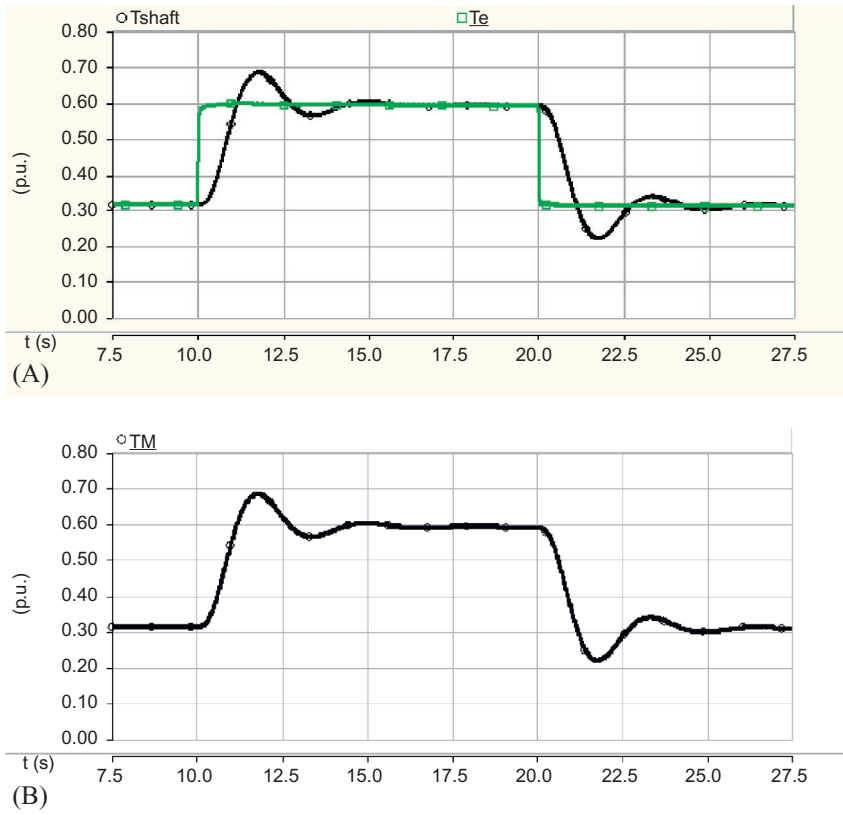


Fig. 4.58 (A) Shaft torque and electric torque and (B) turbine mechanical torque of the microturbine in stand-alone mode.

Remote monitoring, quiet operation, oxygen rich exhaust, and so on are also favorite issues to MTG manufacturers.

In general, manufacturers focus on MT configuration to produce higher overall efficiency and smaller size for customers. MTs will be a desirable, appropriate, and acceptable source for producing power for years to come.

4.14 SUMMARY

Recently, the development of DG has been a main issue for governments, and high efficiency and reliable sources have been considered. MTs are suitable with respect to their potential merits and they are favorable for

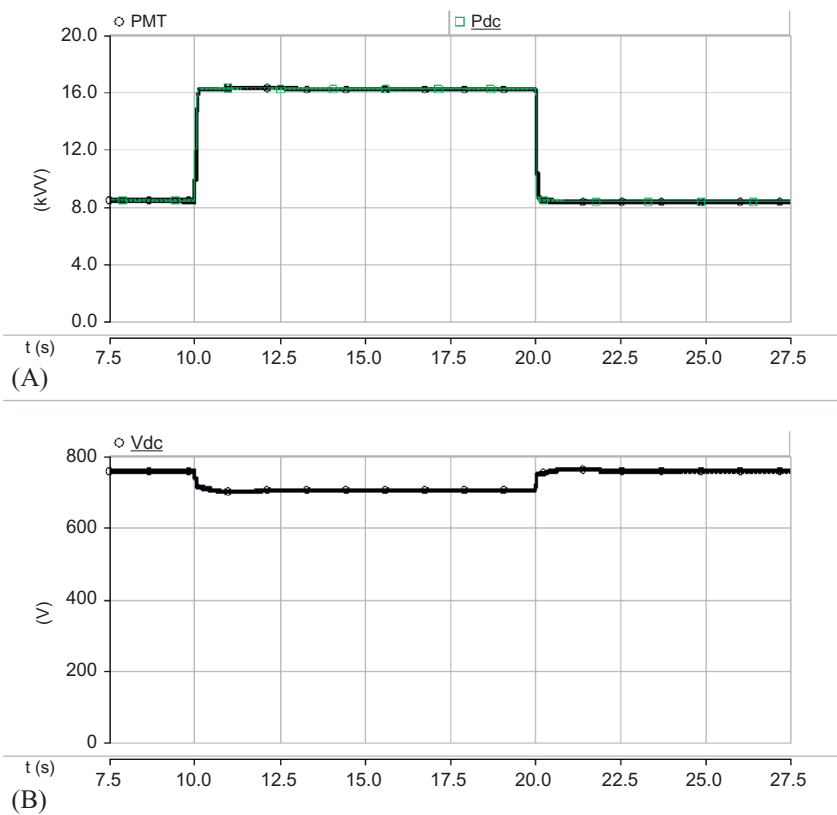


Fig. 4.59 (A) Microturbine and DC bus power output and (B) DC bus voltage in stand-alone mode.

various applications. The MTs' operation is independent of choose of fuel type and they are suitable for CHP. Moreover, start-up time of MTs is short, thus they are serious sources for grid supporting and load supplying.

To begin this chapter, MT components and applications were illustrated and statistical information was presented showing that MTs are a main source for power systems in the future. Modeling of the MT was presented in grid-connected and stand-alone modes and grid-connected control strategy based on $\alpha\beta$ transformation and $dq0$ transformation was demonstrated in islanding mode. Control circuits were described completely. Simulations were carried out in PSCAD/EMTDC software for two mentioned modes. Simulation results approved modeling and the MT performance as a suitable power source.

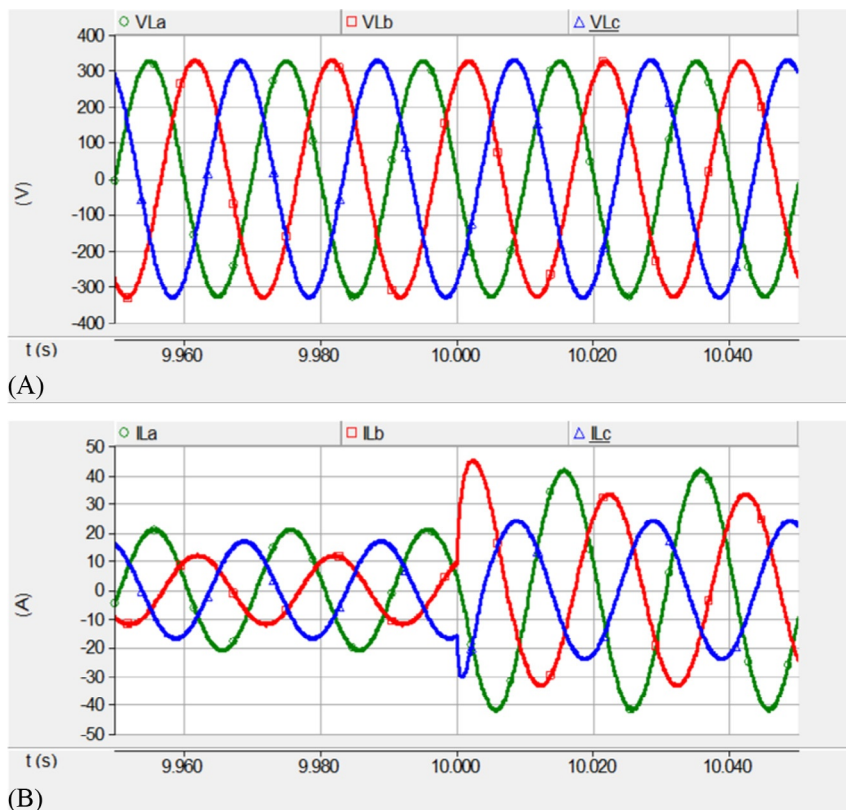


Fig. 4.60 Unbalanced load terminal: (A) phase voltages and (B) line currents.

Finally, in the future MTGs should receive more global attention and enjoy more progress because they are appropriate for the distribution grid and are a clean source of power.

APPENDIX A

Parameters of the MT

- Speed governor parameters:

$W = 25$, $X = 0.4$, $Y = 0.05$, $Z = 1$, speed ref. = 1 p.u., acceleration ref. = 0.01 p.u.

- Fuel system controller parameters:

$$K_6 = 0.23, K_3 = 0.77, a = l, b = 0.05, \\ c = l, T_f = 0.4, E_{CR} = 0.01, E_{TD} = 0.04, \\ Max\ limit = 1.5 \text{ p.u.}, Min\ limit = -0.1 \text{ p.u.}$$

- Compressor–turbine combination parameters:

$$T_{CD} = 0.2, a_{f2} = -0.3, b_{f2} = 1.3, c_{f2} = 0.5$$

- Temperature controller parameters:

$$K_4 = 0.8, K_5 = 0.2, T_3 = 15, T_4 = 2.5, T_R = 950^\circ\text{F}, T_5 = 3.3, T_t \\ = 450^\circ\text{F}, a_{f1} = 700, b_{f1} = 550$$

Parameters of the PMSG

Rated output = 30 kW, Rated voltage = 0.320 kV, Base Angular Frequency = 10,053.1 rad/s, $R_s = 0.25 \Omega$, $p = 2$, $L_d = L_q = 0.0687 \mu\text{H}$, $\lambda = 0.0534 \text{ Wb}$, $J = 0.011 \text{ kg m}^2$, $F = 0$.

Parameters of the Unbalanced AC Loads

- RL load No. 1:

$$z_{la1} = 15.62 \angle 11.61^\circ \Omega, z_{lb1} = 27.48 \angle 6.57^\circ \Omega, z_{lc1} = 19.56 \angle 9.25^\circ \Omega$$

- RL load No. 2:

$$z_{la2} = 7.86 \angle 11.53^\circ \Omega, z_{lb2} = 9.83 \angle 9.21^\circ \Omega, Z_{lc2} = 13.79 \angle 6.54^\circ \Omega$$

APPENDIX B

Ideal gas properties of air [8]

T (K)	h (kJ/kg)	Pr	T (K)	h (kJ/kg)	Pr	T (K)	h (kJ/kg)	Pr
200	199.97	0.3363	510	513.32	9.031	1000	1046	114.0
210	209.97	0.3987	520	523.63	9.684	1020	1068.89	123.4
220	219.97	0.4690	530	533.98	10.37	1040	1091.85	133.3
230	230.02	0.5477	540	544.35	11.10	1060	1114.86	143.9
240	240.02	0.6355	550	555.74	11.86	1080	1137.89	155.2
250	250.05	0.7329	560	565.17	12.66	1100	1161.07	167.1
260	260.09	0.8405	570	575.59	13.50	1120	1184.28	179.7
270	270.11	0.9590	580	586.04	14.38	1140	1207.57	193.1
280	280.13	1.0889	590	596.52	15.31	1160	1230.92	207.2
285	285.14	1.1584	600	607.02	16.28	1180	1254.34	222.2
290	290.16	1.2311	610	617.53	17.30	1200	1277.79	238.0
295	295.17	1.3068	620	628.07	18.36	1220	1301.31	254.7
298	298.18	1.3543	630	638.07	18.36	1240	1324.93	272.3
300	300.19	1.3860	640	649.22	20.64	1260	1348.55	290.8
305	305.22	1.4686	650	659.84	21.86	1280	1372.24	310.4
310	310.24	1.5546	660	670.47	23.13	1300	1395.97	330.9
315	315.27	1.6442	670	681.14	24.46	1320	1419.76	352.5
320	320.29	1.7375	680	691.82	25.85	1340	1443.60	375.3
325	325.31	1.8345	690	702.52	27.29	1360	1467.49	399.1
330	330.34	1.9352	700	713.27	28.80	1380	1491.44	424.2
340	340.42	2.149	710	724.04	30.38	1400	1515.42	450.5
350	350.49	2.379	720	734.82	32.02	1420	1539.44	478.0
360	360.58	2.626	730	745.62	33.72	1440	1563.51	506.9
370	370.67	2.892	740	756.44	35.50	1460	1587.63	537.1
380	380.77	3.176	750	767.29	37.35	1480	1611.79	568.8
390	390.88	3.481	760	778.18	39.27	1500	1635.97	601.9
400	400.98	3.806	780	800.03	43.35	1520	1660.23	636.5
410	411.12	4.153	800	821.95	47.75	1540	1684.51	672.8
420	421.26	4.522	820	843.98	52.59	1560	1708.82	710.5
430	431.43	4.915	840	866.08	57.60	1580	1733.17	750.0
440	441.61	5.332	860	888.27	63.09	1600	1757.57	791.2
450	451.80	5.775	880	910.56	68.98			
460	462.02	6.245	900	932.93	75.29			
470	472.24	6.742	920	955.38	82.05			
480	482.49	7.268	940	977.92	89.28			
490	492.74	7.824	960	1000.55	97.00			
500	503.02	8.411	980	1023.25	105.2			

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